T2K 6.914×10^{20}-POT Neutrino 4.011×10^{20}-POT Anti-Neutrino 3-Flavour Joint Analysis

Costas Andreopoulos[a,b], Christopher Barry[b], Thomas Dealtry[f], Steve Dennis[a,d], Davide Sgalaberna[g], Raj Shah[a,c]

[a] *Science & Technology Facilities Council (STFC), Rutherford Appleton Laboratory, Harwell Oxford Campus, OX11 0QX, UK*
[b] *University of Liverpool, Physics Department, Liverpool, L69 7ZE, UK*
[c] *University of Oxford, Subdepartment of Particle Physics, Oxford, OX1 3RH, UK*
[d] *University of Warwick, Physics Department, Coventry, CV4 7AL, UK*
[e] *CSIC and University of Valencia, Instituto de Fisica Corpuscular, E-46071 Valencia, Spain*
[f] *Lancaster University, Physics Department, Lancaster, LA1 4YB, UK*
[g] *University of Geneva, Departement de physique nucleaire et corpusculaire, 1211 Geneve, Switzerland*

Abstract

We report the results of the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1+6 dataset, using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for CP violation by setting limits on δ_{CP} and measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) (Δm^2_{32} for normal hierarchy and Δm^2_{13} for inverted hierarchy). Though T2K has limited sensitivity to the mass hierarchy given the relatively short baseline, fits under both hypotheses are performed to show which is preferred by data. Furthermore the search for $\bar{\nu}_e$ appearance and the measurement of $\bar{\nu}_\mu$ disappearance are shown as well.

This analysis adapts the work presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] to this dataset. The Run 1-6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 6.914×10^{20} POT in neutrino mode and 4.011×10^{20} POT in antineutrino mode.

This analysis predicts 125.914 ± 6.459 (syst) for neutrino μ-like, 34.284 ± 1.755 (syst) for antineutrino μ-like, 26.594 ± 1.808 (syst) for neutrino e-like, and 3.218 ± 0.238 (syst) for antineutrino e-like single ring events in Super K, given the oscillation parameters of the Asimov data set A in Tab. 12, and observes respectively 124 for neutrino μ-like, 34 for antineutrino μ-like, 31 for neutrino e-like, 3 for antineutrino e-like.

A joint analysis of all the four selected samples is performed in a framework of 3-flavour oscillations including matter effects in constant-density matter.

For μ-like samples the templates are built as a function of the reconstructed neutrino energy, while for e-like the analysis templates are built as a function of the reconstructed neutrino energy and lepton angle, in order to improve the separation of different flavour/reaction components and increase the sensitivity to δ_{CP}. The observed single ring events spectra are compared to the expected spectra in order to infer all the 3-flavour oscillation parameters.

As in [10], all the systematic parameters are marginalised [55] in the fit and confidence intervals are quoted for all the oscillation parameters. Since the most important result of this analysis release is the search for CP violation, the Feldman-Cousins method [54] is performed only for the parameter δ_{CP}. Since Feldman-Cousins requires a lot of CPU time, it will be computed during the first review of the note and added to a later version.

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1. Introduction

In this technical note, we report the results of a 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1-6 dataset, using data taken in both neutrino mode (6.914×10^{20} POT), with the polarity set to forward horn current (FHC), and antineutrino (4.011×10^{20} POT) mode with polarity set reverse horn current (RHC). This analysis builds on our other analyses presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] As in [10], a hybrid frequentist-Bayesian method [55], which consists of marginalizing instead of profiling the nuisance parameters, is used.

The consistency of the data with the expectation is given by comparing the predicted reconstructed neutrino energy and outgoing lepton angle spectra of 1 ring events at Super-K to the observed spectra. The method of calculation of these predicted spectra is presented in Sec. 2. This analysis uses flux tuning based on recent NA61/SHINE 2009 data [17][46]. It also uses an indirect extrapolation from ND280 to Super-K, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ for different event topologies with the neutrino interaction vertex either in FGD1 or FGD2 [29].

A significant difference between this and the previous analyses is that for e-like samples the analysis templates are built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ and $\nu_\mu \rightarrow \nu_e$, fundamental in measuring δ_{CP}. Furthermore the ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increase the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For μ-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

Updated nominal predictions of the reconstructed neutrino energy and lepton angle spectrum of single ring events in Super-K are presented for Run 1-6 in Sec. 2

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [52]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 101 parameters including energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state re-scattering systematics plus the oscillation parameters (nuisance parameters). In this Run 1-6 analysis, all the nuisance parameters are marginalised [55] . The nuisance parameters and their effects on the extrapolated single ring event spectra are described in Sec. 3. The analysis method is presented in detail in Sec. 4.1

Fit validation and coverage studies are presented in Sec. 5.

The main goal of this analysis is to search for CP violation in the context of the PMNS matrix and confidence intervals are set using the Feldman-Cousins method [54]. Confidence intervals of the other oscillation parameters are set using the constant $\Delta\chi^2$ method. Sensitivity studies are shown in Sec. 5.1. The results of the measurement using the Run 1-6 dataset is given in Sec. 6 of this note.

Finally, a summary is given in Sec. 7.

2. Predictions of single ring event spectra in Super-K

2.1. Methodology

In this analysis a comparison of the reconstructed energy (E_r) and lepton angle (θ) spectrum of single ring events observed at Super-K with the predicted spectrum is used to build the test statistic to quote confidence intervals of the oscillation parameters. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in the past analyses, and a unique θ bin is used. The predicted number $N_{SK;r,\theta}$ of single ring events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} \beta \cdot P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and lepton angle bin θ. The construction of the nominal MC templates is discussed in Sec. 2.2. Both flux-tuning weights based on NA61/SHINE hadron-production data (see Sec. 2.3) and flux and cross-section tuning weights based on ND280 data (see Sec. 2.4) are applied to the nominal MC templates. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, the lepton angle bin θ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 3. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in Sec. 2.5.

2.2. Construction of nominal Super-K Monte Carlo templates

The nominal Super-K single ring MC analysis templates $N^{MC}_{SK;m;r\prime,\theta;t}$ are constructed by applying either the single e-like ring selection cuts or the single μ-like ring selection cuts to the official Super-K MC samples. The selection is exactly the same for neutrino and antineutrino beam mode. In total 4 samples are obtained:

- neutrino mode single μ-like ring sample
- antineutrino mode single μ-like ring sample
- neutrino mode single e-like ring sample
- antineutrino mode single e-like ring sample

Details of the MC samples are given in Sec. 2.2.1. The calculation of the normalization of each MC sample is described in Sec. 2.2.2. The official single μ-like ring selection cuts are listed in Sec. 2.2.3. while the official single e-like ring selection cuts are listed in Sec. 2.2.4. The list of MC templates used in this analysis is shown in Sec. 2.2.5, while the true and reconstructed energy as well as lepton angle binning of the templates is discussed in Sec. 2.2.6.

2.2.1. Input Monte Carlo samples

This analysis uses the official 14a Super-K MC samples. These were generated using the 13a JNUBEAM flux, NEUT 5.3.2, skdetsim v13p90 and the 14b Super-K software [33] with minor tunings. The CCQE model is reweighted from the Spectral Function to a Relative Fermi Gas model with a relativistic Random Phase Approximation on an event by event basis. The NC1γ events are scaled up by a factor of two in accordance with improved calculations of single photon emission in NC interactions [34]. Lastly, a simplified reweighting, as a function of the outgoing pion energy, is used to scale the CCCoherent events from the Rein Sehgal model to the Berger-Sehgal model prediction. Six MC samples were generated for both neutrino and antineutrino beam mode running (horn current $\pm$250kA). Four of the samples ($\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$ and ν_e) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample. The sixth sample is ν_e interactions generated with the ν_μ flux, and will be denoted the "oscillated ν_e" sample.

Running mode	Neutrino type	Number of events (flux averaged) per 22.5 kt fiducial per 10^{21} POT	Number of generated events in true fiducial volume in all available MC
Neutrino	$\bar{\nu}_\mu$	50.964	634600
	ν_μ	1396.245	634881
	$\bar{\nu}_e$	2.884	126728
	ν_e	26.663	126899
	Oscillated $\bar{\nu}_e$	52.939	126984
	Oscillated ν_e	1483.994	126916
Antineutrino	$\bar{\nu}_\mu$	353.789	633988
	ν_μ	208.369	634659
	$\bar{\nu}_e$	6.811	127013
	ν_e	8.941	127262
	Oscillated $\bar{\nu}_e$	376.420	126812
	Oscillated ν_e	215.099	127278

Table 1: Normalization of input Super-K MC samples [33]

2.2.2. Normalization of Monte Carlo samples

The normalization (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{2}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section on water for the given neutrino species, I is the beam intensity in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The Super-K MC samples are normalised to the Run 1-6 POT (6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode) using the numbers in Tab. 1. Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with the ratio of the number of events per 22.5 kt per 10^{21} POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to 6.914×10^{20} POT for neutrino and 4.011×10^{20} for antineutrino mode, by weighting the result respectively by 0.6914 and 0.4011 (the Run 1-6 POT in units of 10^{21} POT).

2.2.3. Single μ-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single μ-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

The same selection is applied to both neutrino and antineutrino mode samples.

2.2.4. Single e-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single e-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts
- the reconstructed vertex is at least 2 metres away from the ID wall
- the visible energy in the event is above 30 MeV
- there is exactly 1 reconstructed ring in the event
- the reconstructed ring is e-like
- number of decay electrons is zero
- the reconstructed neutrino energy is below 1250 MeV
- fiTQun rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

The same selection is applied to both neutrino and antineutrino mode samples.

2.2.5. List of Monte Carlo templates

For each Super-K MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6].

In our previous $\bar{\nu}_\mu$-disappearance and $\bar{\nu}_e$ appearance analyses the NC interaction modes were subdivided into NC1$\pi^\pm$, NC1π^0, NC coherent and NC other. In this analysis the new mode NC1γ, which was previously included in NC other, is used. We have also added 5 CC MEC templates for $\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$, ν_e and oscillated ν_e and 5 templates for the oscillated $\bar{\nu}_e$ MC sample. This gives a total of 50 templates:

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC MEC,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC 1γ,
- ν_μ CCQE,
- ν_μ CC MEC,
- ν_μ CC1π,
- ν_μ CC coherent,

- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC 1γ,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC MEC,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC 1γ,
- ν_e CCQE,
- ν_e CC MEC,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1π^0,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC 1γ,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC MEC,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC MEC,

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	66.077	0.035	4.291	0.002	0.225	0.001	70.631
CC1pi	16.973	0.017	1.566	0.001	0.067	0.000	18.625
CCcoherent	0.326	0.000	0.080	0.000	0.001	0.000	0.408
MEC	19.038	0.010	0.845	0.000	0.050	0.000	19.944
CCother	7.735	0.008	0.494	0.001	0.002	0.000	8.240
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	117.643	0.297	7.602	0.025	0.345	0.002	125.914

Table 2: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set A in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	6.993	0.004	12.792	0.005	0.006	0.014	19.813
CC1pi	2.707	0.003	3.033	0.002	0.002	0.003	5.750
CCcoherent	0.054	0.000	0.169	0.000	0.000	0.000	0.223
MEC	2.503	0.001	2.053	0.001	0.002	0.002	4.562
CCother	1.280	0.001	0.885	0.000	0.000	0.000	2.168
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.508	0.051	19.664	0.032	0.010	0.019	34.284

Table 3: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set A in tab. 12 is used.

- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC1π,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC other.

In Tab. 2, 3, 4 and 5 the predicted number of events is shown for all the selected samples and each flavour and reaction mode.

2.2.6. Binning of Monte Carlo templates

This analyses uses an updated binning scheme for reconstructed variables. The difference with previous analyses is that for e-like samples the analysis templates are built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring δ_{CP}. Furthermore the ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increases the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For μ-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

μ-like templates have 84 true energy bins and 73 reconstructed energy bins. e-like templates have 84 true energy bins, 25 reconstructed energy bins and 15 lepton angle bins.

For all templates the 84 true energy bins of constant width are the following:

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.056	2.129	0.001	0.088	15.875	0.099	18.249
CC1pi	0.016	0.348	0.000	0.027	1.688	0.017	2.096
CCcoherent	0.000	0.002	0.000	0.002	0.015	0.002	0.021
MEC	0.008	0.710	0.000	0.022	3.901	0.019	4.659
CCother	0.001	0.045	0.000	0.003	0.029	0.001	0.080
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.469	3.264	0.069	0.145	21.508	0.138	26.594

Table 4: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set A in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.375	1.156	2.065
CC1pi	0.002	0.040	0.001	0.060	0.057	0.137	0.298
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.019	0.026
MEC	0.001	0.069	0.000	0.063	0.114	0.185	0.433
CCother	0.001	0.008	0.000	0.004	0.004	0.002	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.141	0.313	0.235	0.478	0.551	1.500	3.218

Table 5: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set A in tab. 12 is used.

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 50-MeV bins from 0-1.25 GeV.

which is identical to the past analyses.

For μ-like samples the 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,

- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

The θ binning used for e-like samples is

- 14 bins with constant width from 0° to 140°,
- 1 bin for the range $140° - 180°$.

that is the same as used by the $p - \theta$ analysis group [15].

For μ-like samples a unique θ bin, that covers the whole range 0° - 180°, is used. The same reconstructed neutrino energy and lepton angle binning is used for the fit of the single ring reconstructed energy spectra. The predicted e-like 2-dimensional spectra are shown in App. B for each flavour component and compared with the 1-dimensional reconstructed energy spectra. It is clear that the components $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ are better separated by the reconstructed energy Vs θ templates.

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4]. The binning described here is unchanged with respect to the Run 1+2+3+4 joint fit analysis.

2.3. Applying flux tuning based on NA61/SHINE data

In this analysis, the nominal MC templates, which are constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which is tuned on NA61/SHINE hadron production measurements [36][45]. The reweighted MC templates are denoted the *NA61/SHINE-tuned* MC templates in this document. The applied weights are obtained from [36] and are shown as a function of the true neutrino energy in Figs. 1-4. The effect of this flux tuning on the predicted spectra events is discussed in Sec. 2.6

2.4. Applying flux and cross-section (BANFF) tuning based on ND280 data

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples to find the best-fit values of the flux normalisation and cross-section parameters [28]. This analysis uses these best-fit values to reweight the NA61/SHINE-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their 1σ prefit and postfit errors is given in Tab. 6, 7, 8 and 9.

The effects of the BANFF tune on the predicted spectra is discussed in Sec. 2.6.

2.5. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\overset{(-)}{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}. The $\overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_e \to \overset{(-)}{\nu}_e)$, while the oscillated $\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e)$ (the sample was generated assuming 100% of $\overset{(-)}{\nu}_\mu$ transform to $\overset{(-)}{\nu}_e$). All these oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code are first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [69], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of $\overset{(-)}{\nu}_e$ and $\overset{(-)}{\nu}_\mu$ can yield $\overset{(-)}{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect

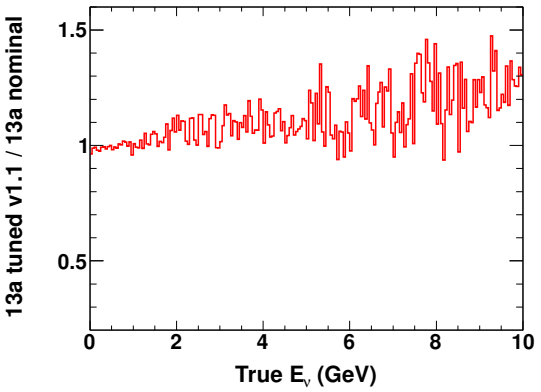


Figure 1: ν_μ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [36].

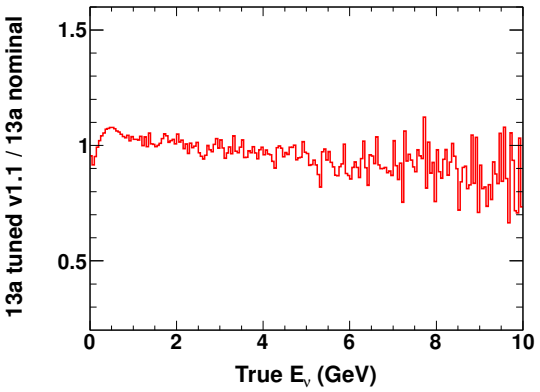


Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [36].

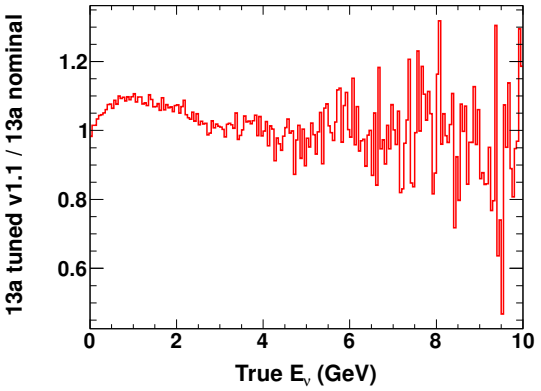


Figure 3: ν_e 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_e MC templates [36].

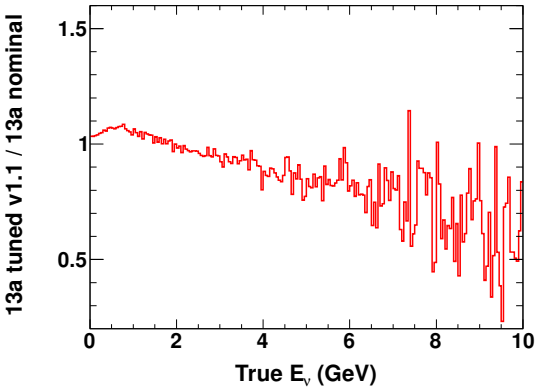


Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [36].

is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ Super-K MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m are proxies for the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

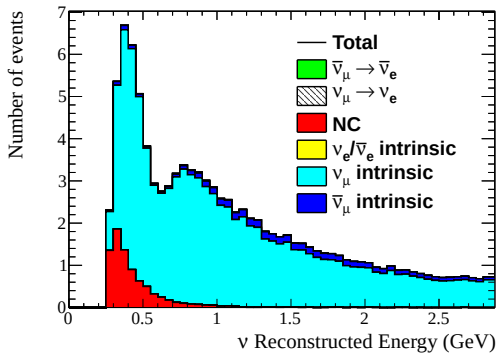
Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

2.6. Predictions of nominal and tuned-MC spectra

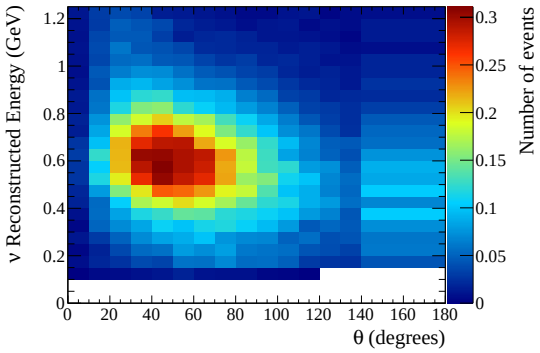
In this section, we present the expected spectra in Super-K for various oscillation scenarios. Unless explicitly stated otherwise, the oscillation parameters are set to the values shown in Tab. 12 and the normal mass hierarchy is assumed. All plots are generated for an integrated exposure of 6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode.

Predicted single ring Super-K spectra are shown in Fig. 5. By comparing neutrino and antineutrino mode e-like plots it is clear that E_{reco} Vs θ distributions are differently populated by $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events. More details on the advantage of the 2-dimensional templates are given in App. B.

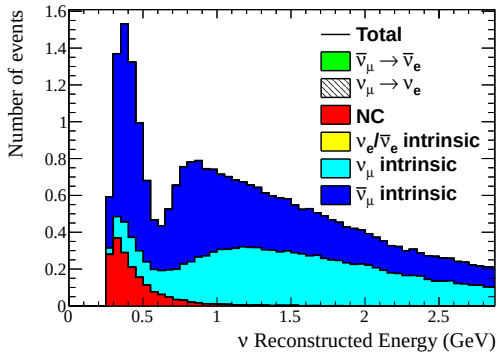
[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like Super-K sample. A Super-K MC study is required to determine whether the Super-K



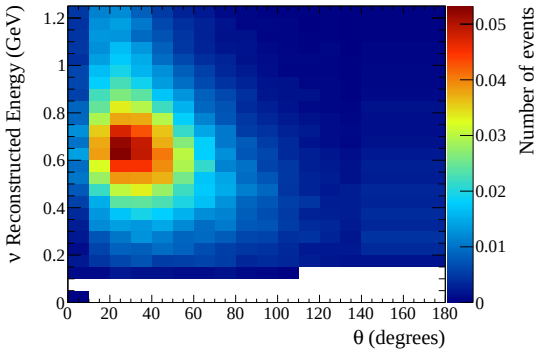
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode

Figure 5: Predicted spectra, muon-like distributions are function of the reconstructed neutrino energy, while the electron-like distributions are function of both the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode and of 4.011×10^{20} POT in antineutrino mode. The muon-like distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra are generated with the systematic parameters measured with the ND280 data shown in Tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

efficiency assumptions made there are valid.

3. Systematic effects in the prediction of the single μ-like ring event energy spectrum

3.1. Input systematic parameters

This analysis considers 101 systematic parameters which can be grouped into 4 categories:

1. **Flux and cross-section parameters** (65 parameters).

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit. However, since ND280 is not sensitive to some cross section systematics they remain unconstrained (detailed later). Parameters fit are 11 ν_μ, 5 $\bar{\nu}_\mu$, 7 ν_e and 2 $\bar{\nu}_e$ neutrino mode (FHC) flux parameters, 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ antineutrino mode (RHC) flux parameters and 15 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples as described in Ref [28]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 6, 7, 8 and 9. The best-fit values are used to reweight the templates as described in Sec. 2.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis.

The effects of these correlated uncertainties are evaluated using the Cholesky method, that allows to throw the systematic parameters according to their multivariate distribution taking into account the existing correlations.

Some of the cross-section parameters have a complicated, often non-linear, effects on true and reconstructed neutrino energy bin normalisations; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3σ to +3σ in steps of 0.5σ, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7σ to +3σ). Cubic splines are made for the following cross-section uncertainties:

- CCQE axial-mass scaling factor (MaQE)
- Fermi momentum for ^{16}O
- Binding energy for ^{16}O
- Nucleon to Δ transition axial form factor (C_5^A)
- Resonance-production axial-mass scaling factor (MaRes)
- Scale of isospin 1/2 nonresonant background
- CC other shape

The splines are made in bins of theta, true and reconstructed energy, and the binning is given in Ref. [35].

The effects of the flux and cross-section normalization uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalization for a limited range of true energy). The cross section parameters which simply normalise the overall spectrum component are:

- MEC, applied to all the MEC events (both neutrino and antineutrino)
- MEC-$\bar{\nu}$, applied to all the antineutrino MEC events
- CC coherent for ^{16}O normalisation
- NC coherent normalisation
- NC other normalisation
- NC 1γ normalisation
- CC ν_e normalisation
- CC $\bar{\nu}_e$ normalisation

Uncertainties due to "second class currents" are implemented as an additional uncertainty on CC $\bar{\nu}_e$ and CC ν_e parameters with a 2% anti-correlated constraint between them. The BANFF parameters are applied to the MC templates as specified in Tab. 34, 35, 36 and 37. The normalisation parameter of the NC1γ component was not used in the previous analyses, since it was included in NC other.

In the BANFF fit a bias on the best-fit flux parameters is present due to the degeneracy between the neutrino flux and ND280 detector parameters. For this reason a covariance matrix that accounts for the bias of the flux as well as all the other parameters is summed in quadrature to the total post-fit covariance. Furthermore the best-fit values have been corrected in order to remove the bias. The description of the procedure is given in [28]. As shown in section D the impact of the bias on the systematic parameters fitted by the BANFF analysis is small compared to the other systematic uncertainties.

The predicted number of events in antineutrino mode e-like sample is now smaller compared to the $\bar{\nu}_e$ appearance analysis presented at the 2015 summer conferences [10], 3.218 instead of 3.729. The differences between the current and previous oscillation analyses is mainly due to:

(a) new value of $\sin^2\theta_{13}$ corresponding to the PDG 2015 release, as shown in Tab. 12, which is 0.0217 while in the previous analysis it was 0.0250. If the old PDG value is used the expected number of antineutrino mode e-like events becomes 3.542.

(b) Subsequently, if the cross section parameters are set to the old post-BANFF fit values, the expected number of events becomes 3.63. Indeed all the post-BANFF 2015 fit cross section parameters are higher compared to the 2016 BANFF fit result. The effect of the flux parameters is smaller, indeed if they are set to BANFF 2015 result, the expected number of events becomes 3.66.

(c) Finally, after all the changes listed above, the rescaling of CC coherent and NC1γ components (see 2.2.1) is not applied, the agreement between the 2015 and 2016 analysis is obtained. The first one increases the number of events up to 3.772 while the latter one reduces down to 3.711, which is approximately the number of events expected in the 2015 analysis.

Parameter	Description	Best fit	1σ pre/postfit fractional error
0	FHC ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.128	0.099 / 0.064
1	FHC ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.156	0.103 / 0.061
2	FHC ν_μ flux normalisation, E = 0.5 - 0.6 GeV	1.148	0.096 / 0.051
3	FHC ν_μ flux normalisation, E = 0.6 - 0.7 GeV	1.128	0.087 / 0.043
4	FHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.104	0.113 / 0.046
5	FHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.100	0.092 / 0.045
6	FHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.127	0.070 / 0.044
7	FHC ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.124	0.074 / 0.048
8	FHC ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.121	0.087 / 0.049
9	FHC ν_μ flux normalisation, E = 5.0 - 7.0 GeV	1.075	0.098 / 0.053
10	FHC ν_μ flux normalisation, E = 7.0 - 30.0 GeV	1.064	0.114 / 0.065
11	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.7 GeV	1.100	0.103 / 0.081
12	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.112	0.079 / 0.048
13	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.111	0.084 / 0.060
14	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.116	0.086 / 0.070
15	FHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 30.0 GeV	1.162	0.086 / 0.069
16	FHC ν_e flux normalisation, E = 0.0 - 0.5 GeV	1.134	0.090 / 0.052
17	FHC ν_e flux normalisation, E = 0.5 - 0.7 GeV	1.135	0.090 / 0.049
18	FHC ν_e flux normalisation, E = 0.7 - 0.8 GeV	1.135	0.086 / 0.047
19	FHC ν_e flux normalisation, E = 0.8 - 1.5 GeV	1.119	0.081 / 0.043
20	FHC ν_e flux normalisation, E = 1.5 - 2.5 GeV	1.115	0.079 / 0.046
21	FHC ν_e flux normalisation, E = 2.5 - 4.0 GeV	1.111	0.084 / 0.050
22	FHC ν_e flux normalisation, E = 4.0 - 30.0 GeV	1.118	0.094 / 0.067
23	FHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 2.5 GeV	1.121	0.074 / 0.057
24	FHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 30.0 GeV	1.153	0.128 / 0.117

Table 6: Summary of neutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Parameter	Description	Best fit	1σ fractional error
25	RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	1.098	0.094 / 0.072
26	RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.121	0.079 / 0.052
27	RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.130	0.077 / 0.048
28	RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.155	0.081 / 0.054
29	RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.111	0.080 / 0.055
30	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	1.118	0.104 / 0.071
31	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	1.127	0.102 / 0.060
32	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	1.117	0.096 / 0.052
33	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	1.121	0.085 / 0.044
34	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.155	0.125 / 0.066
35	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.132	0.105 / 0.057
36	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.139	0.080 / 0.053
37	RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	1.141	0.074 / 0.054
38	RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	1.151	0.094 / 0.071
39	RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	1.133	0.093 / 0.070
40	RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	1.082	0.130 / 0.110
41	RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.117	0.069 / 0.051
42	RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.112	0.085 / 0.071
43	RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	1.126	0.095 / 0.058
44	RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	1.127	0.091 / 0.051
45	RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	1.133	0.091 / 0.052
46	RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	1.132	0.084 / 0.046
47	RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	1.125	0.080 / 0.056
48	RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	1.119	0.089 / 0.071
49	RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	1.166	0.156 / 0.141

Table 7: Summary of antineutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Parameter	Description	Best fit	1σ pre/postfit fractional error
50	Meson exchange current normalisation for ^{16}O	1.546	- / 0.343
51	C_5^A nucleon to Δ transition axial form factor	0.789	0.119 / 0.062
52	Scale of isospin 1/2 nonresonant background	1.042	0.154 / 0.134
53	CCQE axial-mass scaling factor	0.931	- / 0.028
54	Resonance-production axial-mass scaling factor	0.889	0.158 / 0.041
55	BANFF; Second current class axial	1.000	1.000 / 1.000
56	BANFF; Second current class vector	1.000	1.000 / 1.000
57	Fermi momentum for ^{16}O	1.041	- / 0.105
58	CC other shape	-0.022	0.400 / 0.208
59	Binding energy for ^{16}O	0.882	0.333 / 0.282
60	CC coherent for ^{16}O normalisation	0.858	0.300 / 0.228
61	NC coherent normalisation	0.931	0.300 / 0.298
62	NC other normalisation	1.000	0.300 / 0.300
63	CC ν_e normalisation	1.000	0.028 / 0.028
64	NC 1γ normalisation	1.000	1.000 / 1.000
65	CC $\bar{\nu}_e$ normalisation	1.000	0.028 / 0.028
66	Antineutrino Meson exchange current normalisation for ^{16}O	0.578	- / 0.177

Table 8: Summary of cross section systematics included in the *VALOR* joint fit analysis. Parameters with no prefit error were unconstrained. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Parameter	Description	1σ fractional error
0	SKDet + FSI/SI 0; E_{reco} 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
1	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.017
2	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
3	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu)	0.077
4	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu)	1.004
5	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu)	0.598
6	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re)	0.100
7	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re)	0.030
8	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re)	0.033
9	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.225
10	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.166
11	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.028
12	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.063
13	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.040
14	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.061
15	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re)	0.488
16	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re)	0.194
17	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re)	0.590
18	SKDet + FSI/SI 0; Ereco 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.018
19	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.016
20	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$CCQE (1Rmu); RHC	0.016
21	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu); RHC	0.093
22	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu); RHC	1.004
23	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu); RHC	0.597
24	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re); RHC	0.060
25	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re); RHC	0.027
26	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re); RHC	0.056
27	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.244
28	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.123
29	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	0.904
30	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.044
31	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.032
32	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.068
33	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re); RHC	0.482
34	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re); RHC	0.171
35	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re); RHC	0.538

Table 9: Summary of SK detector + FSI +SI + PN systematics included in the *VALOR* joint fit analysis.

Parameter	Sample	Category	Energy range (GeV)
0 | 18	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
1 | 19	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
2 | 20	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
3 | 21	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$E_{reco} < 0.35$
4 | 22	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.35 < E_{reco} < 0.80$
5 | 23	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.80 < E_{reco} < 1.25$
6 | 24	electron-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
7 | 25	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
8 | 26	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
9 | 27	electron-like	all NC	$E_{reco} < 0.35$
10 | 28	electron-like	all NC	$0.35 < E_{reco} < 0.80$
11 | 29	electron-like	all NC	$0.80 < E_{reco} < 1.25$
12 | 30	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$E_{reco} < 0.40$
13 | 31	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$0.40 < E_{reco} < 1.10$
14 | 32	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$1.10 < E_{reco} < 30.0$
15 | 33	muon-like	all $\nu_\mu/\bar{\nu}_\mu$ CCnQE	$E_{reco} < 30.0$
16 | 34	muon-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 30.0$
17 | 35	muon-like	all NC	$E_{reco} < 30.0$

Table 10: Binning of Super-K selection efficiency systematic parameters in covariance matrices for both electron- and muon-like samples for each category and reconstructed neutrino energy range. The same parametrization is used for both neutrino and antineutrino mode samples. The parameter number is shown for both neutrino (0-17) and antineutrino (18-35) mode.

2. **Super-K efficiencies & effects of intranuclear and secondary re-interactions** (36 parameters).

The Super-K detector uncertainties include the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples. The NC PID error is evaluated using a NC-enriched control sample, which does not use PID information. Each Monte Carlo control sample is fitted to the data using the efficiency as a fit parameter, and the uncertainties in the efficiencies are estimated by adding the covariance matrix containing the differences between the best-fit and nominal efficiencies to the covariance matrix containing the statistical errors in the fitted efficiencies [30] [31].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [20]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [20]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the Super-K detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [23]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single e-like or μ-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [19]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [20]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the Super-K detector efficiency and FSI+SI systematics together, and add their covari-

Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	gauss	0.085 ± 0.005
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) T2K only	uniform	$[0; 0.4]$
$\sin^2 2\theta_{12}$	gauss	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	uniform	$[2; 3] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	gauss	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	uniform	$[-\pi; +\pi]$
Mass Hierarchy	fixed	NH or IH

Table 11: Treatment of the oscillation parameters in the $\nu/\bar{\nu}$ joint analysis. All the gaussian priors are from [66]. The parameter $\sin^2 2\theta_{13}$ can be constrained with the measurement of reactor experiments using the gaussian prior, otherwise a uniform prior is used and the measurement relies only on the T2K data. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [66], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH.

ance matrices linearly. Each of these covariance matrices have 36 correlated parameters, corresponding to different modes and reconstructed energy bins, as shown in Tab. 10. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **Super-K energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the Super-K reconstructed energy scale. This uncertainty is estimated to be 2.4% [32]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in Tab. 34, 35, 36 and 37 and is common to all the super-K samples.

4. **Oscillation parameters and mass hierarchy** (7 parameters).

The 3-flavour oscillation probability is used. All the nuisance oscillation parameters, i.e. those parameters which are not shown in a given contour, are marginalised except the mass hierarchy which is fixed to either normal or inverted hierarchy. The treatment of the oscillation parameters in the analysis is shown in Tab. 11 while the nominal values are shown in Tab. 12. For $\sin^2\theta_{13}$ two different priors can be used, depending on whether the reactor constraint is applied, denoted as "reactors", or not, denoted as "T2K only". In the first case a gaussian prior given by the combined measurement of reactor experiments is used, while in the latter case a flat prior which covers a large range of values is adopted. With respect to the older global fit [65], the value of $\sin^2\theta_{13}$ has been reduced of about 10% and the error reduced as well. Since an excess of events was seen in the T2K Run 1-4 neutrino mode e-like sample and not much data are collected in neutrino mode during the Run 5+6, this change is expected to have an impact on the search for CP violation, making the confidence intervals tighter.

Parameter(s)	Asimov A	Asimov B
$\sin^2\theta_{23}$	0.528	0.45
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	0.085 (0.0217)	0.085 (0.0217)
$\sin^2 2\theta_{12}$	0.846	0.846
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601	0
Mass Hierarchy	Normal	Normal

Table 12: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [66], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH. The only difference between the Asimov data sets A and B is given by δ_{CP}, set to 0, and $\sin^2\theta_{23}$ which corresponds to the opposite octant.

3.2. Evaluation of the effects of systematic parameters on the predicted single e-like ring reconstructed energy spectrum

In this section, we present the effects of the systematics described in Sec. 3.1 on the predicted number of single ring events and the predicted spectra in Super-K. All results are generated for an integrated exposure of 6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode and the BANFF-tuned MC templates are used.

In Tab. 13, 14, 15 and 16 the effect of each single systematic parameter obtained by varying its value by $\pm1\sigma$ is shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	120.349	125.914	126.429	0.40884 %	125.399	-0.40884 %
$f^{banff}_{1;t,r}$	120.349	125.914	126.409	0.39338 %	125.418	-0.39338 %
$f^{banff}_{2;t,r}$	120.349	125.914	126.033	0.094558 %	125.795	-0.094558 %
$f^{banff}_{3;t,r}$	120.349	125.914	126.012	0.078072 %	125.815	-0.078072 %
$f^{banff}_{4;t,r}$	120.349	125.914	126.834	0.73082 %	124.994	-0.73082 %
$f^{banff}_{5;t,r}$	120.349	125.914	126.771	0.68095 %	125.056	-0.68095 %
$f^{banff}_{6;t,r}$	120.349	125.914	126.733	0.65073 %	125.094	-0.65073 %
$f^{banff}_{7;t,r}$	120.349	125.914	126.452	0.42768 %	125.375	-0.42768 %
$f^{banff}_{8;t,r}$	120.349	125.914	126.471	0.44277 %	125.356	-0.44277 %
$f^{banff}_{9;t,r}$	120.349	125.914	126.126	0.16892 %	125.701	-0.16892 %
$f^{banff}_{10;t,r}$	120.349	125.914	125.96	0.036964 %	125.867	-0.036964 %
$f^{banff}_{11;t,r}$	120.349	125.914	125.941	0.021242 %	125.887	-0.021242 %
$f^{banff}_{12;t,r}$	120.349	125.914	125.934	0.015671 %	125.894	-0.015671 %
$f^{banff}_{13;t,r}$	120.349	125.914	126.002	0.069974 %	125.826	-0.069974 %
$f^{banff}_{14;t,r}$	120.349	125.914	126.08	0.13209 %	125.747	-0.13209 %
$f^{banff}_{15;t,r}$	120.349	125.914	126.063	0.1185 %	125.765	-0.1185 %
$f^{banff}_{16;t,r}$	120.349	125.914	125.914	0.00034323 %	125.913	-0.00034323 %
$f^{banff}_{17;t,r}$	120.349	125.914	125.915	0.00058236 %	125.913	-0.00058236 %
$f^{banff}_{18;t,r}$	120.349	125.914	125.914	0.00041762 %	125.913	-0.00041762 %
$f^{banff}_{19;t,r}$	120.349	125.914	125.916	0.0021262 %	125.911	-0.0021262 %
$f^{banff}_{20;t,r}$	120.349	125.914	125.917	0.0028071 %	125.91	-0.0028071 %
$f^{banff}_{21;t,r}$	120.349	125.914	125.917	0.0024707 %	125.911	-0.0024707 %
$f^{banff}_{22;t,r}$	120.349	125.914	125.916	0.0015978 %	125.912	-0.0015978 %
$f^{banff}_{23;t,r}$	120.349	125.914	125.914	0.00041569 %	125.913	-0.00041569 %
$f^{banff}_{24;t,r}$	120.349	125.914	125.915	0.0011942 %	125.912	-0.0011942 %
$f^{banff}_{Norm_{MEC}}$	120.349	125.914	130.333	3.5095 %	121.495	-3.5095 %
$f^{banff}_{C_5^A}$	120.349	125.914	127.621	1.3562 %	124.311	-1.2726 %
$f^{banff}_{Bg_{RES}}$	120.349	125.914	126.985	0.85109 %	124.971	-0.74846 %
$f^{banff}_{M_A^{QE}}$	120.349	125.914	127.832	1.5237 %	123.975	-1.5395 %
$f^{banff}_{M_A^{RES}}$	120.349	125.914	127.008	0.86933 %	124.837	-0.85554 %
$f^{banff}_{p_f}$	120.349	125.914	123.763	-1.7078 %	127.899	1.5763 %
$f^{banff}_{Shape_{CCoth}}$	120.349	125.914	126.435	0.41379 %	125.393	-0.41379 %
$f^{banff}_{E_B}$	120.349	125.914	125.899	-0.01212 %	125.93	0.012715 %
$f^{banff}_{Norm_{CCcoh}}$	120.349	125.914	126.022	0.086181 %	125.805	-0.086181 %
$f^{banff}_{Norm_{NCcoh}}$	120.349	125.914	125.919	0.0044519 %	125.908	-0.0044519 %
$f^{banff}_{Norm_{NCother}}$	120.349	125.914	126.901	0.78393 %	124.927	-0.78393 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	120.349	125.914	125.926	0.0093354 %	125.902	-0.0093354 %
$f^{banff}_{Norm_{NC1\gamma}}$	120.349	125.914	125.918	0.0029615 %	125.91	-0.0029615 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	120.349	125.914	125.914	0.00011837 %	125.914	-0.00011837 %
$f^{banff}_{Norm_{MECBar}}$	120.349	125.914	126.172	0.20522 %	125.655	-0.20522 %
$f^{SK}_{E;r}$	120.349	125.914	125.914	-4.5145e-14 %	125.914	-7.9003e-14 %
$f^{SK+FSI}_{0;t,r}$	120.349	125.914	126.074	0.1271 %	125.754	-0.1271 %
$f^{SK+FSI}_{1;t,r}$	120.349	125.914	126.556	0.5104 %	125.271	-0.5104 %
$f^{SK+FSI}_{2;t,r}$	120.349	125.914	126.675	0.60427 %	125.153	-0.60427 %
$f^{SK+FSI}_{3;t,r}$	120.349	125.914	128.014	1.6677 %	123.814	-1.6677 %
$f^{SK+FSI}_{4;t,r}$	120.349	125.914	126.336	0.33551 %	125.491	-0.33551 %
$f^{SK+FSI}_{5;t,r}$	120.349	125.914	130.738	3.8311 %	121.09	-3.8311 %

Table 13: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 6.914×10^{20} POT for neutrino mode.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	24.2563	26.5935	26.6217	0.10583 %	26.5654	-0.10583 %
$f^{banff}_{1;t,r}$	24.2563	26.5935	26.7186	0.47018 %	26.4685	-0.47018 %
$f^{banff}_{2;t,r}$	24.2563	26.5935	26.8418	0.93345 %	26.3453	-0.93345 %
$f^{banff}_{3;t,r}$	24.2563	26.5935	26.8365	0.91353 %	26.3506	-0.91353 %
$f^{banff}_{4;t,r}$	24.2563	26.5935	26.8768	1.0652 %	26.3102	-1.0652 %
$f^{banff}_{5;t,r}$	24.2563	26.5935	26.6316	0.14302 %	26.5555	-0.14302 %
$f^{banff}_{6;t,r}$	24.2563	26.5935	26.6004	0.025826 %	26.5867	-0.025826 %
$f^{banff}_{7;t,r}$	24.2563	26.5935	26.5978	0.015909 %	26.5893	-0.015909 %
$f^{banff}_{8;t,r}$	24.2563	26.5935	26.5992	0.021436 %	26.5878	-0.021436 %
$f^{banff}_{9;t,r}$	24.2563	26.5935	26.5962	0.010031 %	26.5909	-0.010031 %
$f^{banff}_{10;t,r}$	24.2563	26.5935	26.595	0.0054078 %	26.5921	-0.0054078 %
$f^{banff}_{11;t,r}$	24.2563	26.5935	26.5975	0.014836 %	26.5896	-0.014836 %
$f^{banff}_{12;t,r}$	24.2563	26.5935	26.5963	0.010428 %	26.5907	-0.010428 %
$f^{banff}_{13;t,r}$	24.2563	26.5935	26.596	0.0093637 %	26.591	-0.0093637 %
$f^{banff}_{14;t,r}$	24.2563	26.5935	26.595	0.00542 %	26.5921	-0.00542 %
$f^{banff}_{15;t,r}$	24.2563	26.5935	26.5948	0.0046444 %	26.5923	-0.0046444 %
$f^{banff}_{16;t,r}$	24.2563	26.5935	26.6185	0.093913 %	26.5685	-0.093913 %
$f^{banff}_{17;t,r}$	24.2563	26.5935	26.628	0.12954 %	26.5591	-0.12954 %
$f^{banff}_{18;t,r}$	24.2563	26.5935	26.6109	0.065228 %	26.5762	-0.065228 %
$f^{banff}_{19;t,r}$	24.2563	26.5935	26.6464	0.19869 %	26.5407	-0.19869 %
$f^{banff}_{20;t,r}$	24.2563	26.5935	26.598	0.016936 %	26.589	-0.016936 %
$f^{banff}_{21;t,r}$	24.2563	26.5935	26.594	0.0017641 %	26.5931	-0.0017641 %
$f^{banff}_{22;t,r}$	24.2563	26.5935	26.5937	0.00071333 %	26.5933	-0.00071333 %
$f^{banff}_{23;t,r}$	24.2563	26.5935	26.6008	0.027288 %	26.5863	-0.027288 %
$f^{banff}_{24;t,r}$	24.2563	26.5935	26.5937	0.00079831 %	26.5933	-0.00079831 %
$f^{banff}_{Norm_{MEC}}$	24.2563	26.5935	27.6259	3.882 %	25.5612	-3.882 %
$f^{banff}_{C^A_5}$	24.2563	26.5935	26.8014	0.78157 %	26.3972	-0.73812 %
$f^{banff}_{Bg_{RES}}$	24.2563	26.5935	26.6753	0.30745 %	26.5216	-0.27037 %
$f^{banff}_{M^{QE}_A}$	24.2563	26.5935	27.0363	1.6648 %	26.1369	-1.7172 %
$f^{banff}_{M^{RES}_A}$	24.2563	26.5935	26.6925	0.37237 %	26.4929	-0.37839 %
$f^{banff}_{p_f}$	24.2563	26.5935	25.9799	-2.3076 %	27.1501	2.093 %
$f^{banff}_{Shape_{CCoth}}$	24.2563	26.5935	26.6063	0.048094 %	26.5807	-0.048083 %
$f^{banff}_{E_B}$	24.2563	26.5935	26.5533	-0.15134 %	26.626	0.12201 %
$f^{banff}_{Norm_{CCcoh}}$	24.2563	26.5935	26.5991	0.021135 %	26.5879	-0.021135 %
$f^{banff}_{Norm_{NCcoh}}$	24.2563	26.5935	26.6567	0.23764 %	26.5303	-0.23764 %
$f^{banff}_{Norm_{NCother}}$	24.2563	26.5935	26.6352	0.15691 %	26.5518	-0.15691 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	24.2563	26.5935	27.2933	2.6315 %	25.8937	-2.6315 %
$f^{banff}_{Norm_{NC1\gamma}}$	24.2563	26.5935	27.0873	1.8568 %	26.0997	-1.8568 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	24.2563	26.5935	26.6014	0.029817 %	26.5856	-0.029817 %
$f^{banff}_{Norm_{MECBar}}$	24.2563	26.5935	26.6061	0.047159 %	26.581	-0.047159 %
$f^{SK}_{E;r}$	24.2563	26.5935	26.5935	1.3359e-14 %	26.5935	-1.3359e-14 %
$f^{SK+FSI}_{6;t,r}$	24.2563	26.5935	26.8445	0.94359 %	26.3426	-0.94359 %
$f^{SK+FSI}_{7;t,r}$	24.2563	26.5935	27.0622	1.7624 %	26.1248	-1.7624 %
$f^{SK+FSI}_{8;t,r}$	24.2563	26.5935	26.7121	0.44605 %	26.4749	-0.44605 %
$f^{SK+FSI}_{9;t,r}$	24.2563	26.5935	26.6525	0.22178 %	26.5345	-0.22178 %
$f^{SK+FSI}_{10;t,r}$	24.2563	26.5935	26.6244	0.116 %	26.5627	-0.116 %
$f^{SK+FSI}_{11;t,r}$	24.2563	26.5935	26.6021	0.0321 %	26.585	-0.0321 %
$f^{SK+FSI}_{12;t,r}$	24.2563	26.5935	26.6184	0.093385 %	26.5687	-0.093385 %
$f^{SK+FSI}_{13;t,r}$	24.2563	26.5935	26.6657	0.27133 %	26.5214	-0.27133 %
$f^{SK+FSI}_{14;t,r}$	24.2563	26.5935	26.6662	0.27323 %	26.5209	-0.27323 %
$f^{SK+FSI}_{15;t,r}$	24.2563	26.5935	26.8604	1.0036 %	26.3266	-1.0036 %
$f^{SK+FSI}_{16;t,r}$	24.2563	26.5935	26.7358	0.53512 %	26.4512	-0.53512 %
$f^{SK+FSI}_{17;t,r}$	24.2563	26.5935	26.7166	0.46264 %	26.4705	-0.46264 %

Table 14: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 6.914×10^{20} POT for neutrino mode.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}RHC$	33.354	34.2841	34.3212	0.1084 %	34.2469	-0.1084 %
$f^{banff}_{1;t,r}RHC$	33.354	34.2841	34.3316	0.13851 %	34.2366	-0.13851 %
$f^{banff}_{2;t,r}RHC$	33.354	34.2841	34.4224	0.4034 %	34.1458	-0.4034 %
$f^{banff}_{3;t,r}RHC$	33.354	34.2841	34.5137	0.66983 %	34.0544	-0.66983 %
$f^{banff}_{4;t,r}RHC$	33.354	34.2841	34.5214	0.69216 %	34.0468	-0.69216 %
$f^{banff}_{5;t,r}RHC$	33.354	34.2841	34.4278	0.41934 %	34.1403	-0.41934 %
$f^{banff}_{6;t,r}RHC$	33.354	34.2841	34.3655	0.23759 %	34.2026	-0.23759 %
$f^{banff}_{7;t,r}RHC$	33.354	34.2841	34.301	0.049526 %	34.2671	-0.049526 %
$f^{banff}_{8;t,r}RHC$	33.354	34.2841	34.2964	0.036085 %	34.2717	-0.036085 %
$f^{banff}_{9;t,r}RHC$	33.354	34.2841	34.4627	0.52119 %	34.1054	-0.52119 %
$f^{banff}_{10;t,r}RHC$	33.354	34.2841	34.4614	0.51727 %	34.1067	-0.51727 %
$f^{banff}_{11;t,r}RHC$	33.354	34.2841	34.4688	0.53874 %	34.0994	-0.53874 %
$f^{banff}_{12;t,r}RHC$	33.354	34.2841	34.3882	0.30381 %	34.1799	-0.30381 %
$f^{banff}_{13;t,r}RHC$	33.354	34.2841	34.3968	0.32868 %	34.1714	-0.32868 %
$f^{banff}_{14;t,r}RHC$	33.354	34.2841	34.3136	0.086075 %	34.2546	-0.086075 %
$f^{banff}_{15;t,r}RHC$	33.354	34.2841	34.2898	0.016631 %	34.2784	-0.016631 %
$f^{banff}_{16;t,r}RHC$	33.354	34.2841	34.2853	0.0035678 %	34.2828	-0.0035678 %
$f^{banff}_{17;t,r}RHC$	33.354	34.2841	34.2856	0.0045846 %	34.2825	-0.0045846 %
$f^{banff}_{18;t,r}RHC$	33.354	34.2841	34.2841	0.00019776 %	34.284	-0.00019776 %
$f^{banff}_{19;t,r}RHC$	33.354	34.2841	34.2841	0.00023211 %	34.284	-0.00023211 %
$f^{banff}_{20;t,r}RHC$	33.354	34.2841	34.2841	0.00018231 %	34.284	-0.00018231 %
$f^{banff}_{21;t,r}RHC$	33.354	34.2841	34.2843	0.0007317 %	34.2838	-0.0007317 %
$f^{banff}_{22;t,r}RHC$	33.354	34.2841	34.2845	0.0011599 %	34.2837	-0.0011599 %
$f^{banff}_{23;t,r}RHC$	33.354	34.2841	34.2846	0.0015682 %	34.2835	-0.0015682 %
$f^{banff}_{24;t,r}RHC$	33.354	34.2841	34.2846	0.0016844 %	34.2835	-0.0016844 %
$f^{banff}_{Norm_{MEC}}$	33.354	34.2841	35.2949	2.9484 %	33.2732	-2.9484 %
$f^{banff}_{C_5^A}$	33.354	34.2841	34.7665	1.4073 %	33.8496	-1.2673 %
$f^{banff}_{Bg_{RES}}$	33.354	34.2841	34.6114	0.95474 %	33.9962	-0.8396 %
$f^{banff}_{M_A^{QE}}$	33.354	34.2841	34.7782	1.4412 %	33.7964	-1.4226 %
$f^{banff}_{M_A^{RES}}$	33.354	34.2841	34.557	0.79619 %	34.0229	-0.76184 %
$f^{banff}_{p_f}$	33.354	34.2841	33.7369	-1.596 %	34.8188	1.5598 %
$f^{banff}_{Shape_{CCoth}}$	33.354	34.2841	34.4278	0.41939 %	34.1403	-0.41939 %
$f^{banff}_{E_B}$	33.354	34.2841	34.2825	-0.0045403 %	34.2856	0.0043363 %
$f^{banff}_{Norm_{CCcoh}}$	33.354	34.2841	34.3434	0.17311 %	34.2247	-0.17311 %
$f^{banff}_{Norm_{NCcoh}}$	33.354	34.2841	34.2866	0.0073745 %	34.2815	-0.0073745 %
$f^{banff}_{Norm_{NCother}}$	33.354	34.2841	34.5427	0.75445 %	34.0254	-0.75445 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	33.354	34.2841	34.2846	0.0015892 %	34.2835	-0.0015892 %
$f^{banff}_{Norm_{NC1\gamma}}$	33.354	34.2841	34.2844	0.00099164 %	34.2837	-0.00099164 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	33.354	34.2841	34.2848	0.0022231 %	34.2833	-0.0022231 %
$f^{banff}_{Norm_{MECBar}}$	33.354	34.2841	34.9123	1.8323 %	33.6559	-1.8323 %
$f^{SK}_{E;r}$	33.354	34.2841	34.2841	-2.0725e-14 %	34.2841	-2.0725e-14 %
$f^{SK+FSI}_{0;t,r}RHC$	33.354	34.2841	34.3274	0.12641 %	34.2407	-0.12641 %
$f^{SK+FSI}_{1;t,r}RHC$	33.354	34.2841	34.4079	0.36115 %	34.1602	-0.36115 %
$f^{SK+FSI}_{2;t,r}RHC$	33.354	34.2841	34.5168	0.67883 %	34.0513	-0.67883 %
$f^{SK+FSI}_{3;t,r}RHC$	33.354	34.2841	35.0403	2.2058 %	33.5278	-2.2058 %
$f^{SK+FSI}_{4;t,r}RHC$	33.354	34.2841	34.3304	0.13529 %	34.2377	-0.13529 %
$f^{SK+FSI}_{5;t,r}RHC$	33.354	34.2841	35.3403	3.0809 %	33.2278	-3.0809 %

Table 15: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 4.011×10^{20} POT for antineutrino mode.

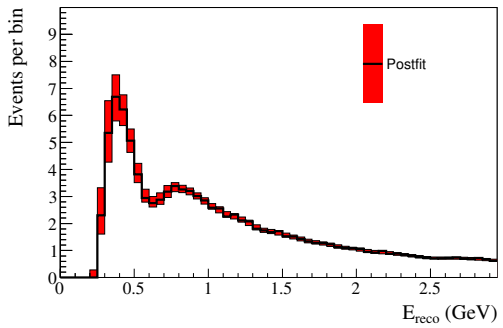
Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}RHC$	3.11707	3.21831	3.23425	0.49537 %	3.20237	-0.49537 %
$f^{banff}_{1;t,r}RHC$	3.11707	3.21831	3.22891	0.32934 %	3.20771	-0.32934 %
$f^{banff}_{2;t,r}RHC$	3.11707	3.21831	3.22364	0.16581 %	3.21297	-0.16581 %
$f^{banff}_{3;t,r}RHC$	3.11707	3.21831	3.22045	0.066577 %	3.21617	-0.066577 %
$f^{banff}_{4;t,r}RHC$	3.11707	3.21831	3.22085	0.078879 %	3.21577	-0.078879 %
$f^{banff}_{5;t,r}RHC$	3.11707	3.21831	3.22043	0.065975 %	3.21619	-0.065975 %
$f^{banff}_{6;t,r}RHC$	3.11707	3.21831	3.22509	0.21083 %	3.21152	-0.21083 %
$f^{banff}_{7;t,r}RHC$	3.11707	3.21831	3.23439	0.49965 %	3.20223	-0.49965 %
$f^{banff}_{8;t,r}RHC$	3.11707	3.21831	3.2365	0.56517 %	3.20012	-0.56517 %
$f^{banff}_{9;t,r}RHC$	3.11707	3.21831	3.25195	1.0453 %	3.18467	-1.0453 %
$f^{banff}_{10;t,r}RHC$	3.11707	3.21831	3.2239	0.17371 %	3.21272	-0.17371 %
$f^{banff}_{11;t,r}RHC$	3.11707	3.21831	3.21966	0.041857 %	3.21696	-0.041857 %
$f^{banff}_{12;t,r}RHC$	3.11707	3.21831	3.21896	0.020249 %	3.21766	-0.020249 %
$f^{banff}_{13;t,r}RHC$	3.11707	3.21831	3.21907	0.023667 %	3.21755	-0.023667 %
$f^{banff}_{14;t,r}RHC$	3.11707	3.21831	3.21857	0.0081707 %	3.21805	-0.0081707 %
$f^{banff}_{15;t,r}RHC$	3.11707	3.21831	3.21855	0.0074537 %	3.21807	-0.0074537 %
$f^{banff}_{16;t,r}RHC$	3.11707	3.21831	3.23255	0.44247 %	3.20407	-0.44247 %
$f^{banff}_{17;t,r}RHC$	3.11707	3.21831	3.21849	0.0055449 %	3.21813	-0.0055449 %
$f^{banff}_{18;t,r}RHC$	3.11707	3.21831	3.22275	0.13805 %	3.21387	-0.13805 %
$f^{banff}_{19;t,r}RHC$	3.11707	3.21831	3.22312	0.14949 %	3.2135	-0.14949 %
$f^{banff}_{20;t,r}RHC$	3.11707	3.21831	3.22095	0.081949 %	3.21567	-0.081949 %
$f^{banff}_{21;t,r}RHC$	3.11707	3.21831	3.22701	0.2705 %	3.2096	-0.2705 %
$f^{banff}_{22;t,r}RHC$	3.11707	3.21831	3.21896	0.020273 %	3.21766	-0.020273 %
$f^{banff}_{23;t,r}RHC$	3.11707	3.21831	3.21839	0.002446 %	3.21823	-0.002446 %
$f^{banff}_{24;t,r}RHC$	3.11707	3.21831	3.21838	0.002219 %	3.21824	-0.002219 %
$f^{banff}_{Norm_{MEC}}$	3.11707	3.21831	3.31425	2.9812 %	3.12237	-2.9812 %
$f^{banff}_{C^A_5}$	3.11707	3.21831	3.25166	1.0364 %	3.19031	-0.86988 %
$f^{banff}_{Bg_{RES}}$	3.11707	3.21831	3.23459	0.5058 %	3.20399	-0.4448 %
$f^{banff}_{M^{QE}_A}$	3.11707	3.21831	3.26276	1.3812 %	3.17411	-1.3732 %
$f^{banff}_{M^{RES}_A}$	3.11707	3.21831	3.22955	0.34945 %	3.20849	-0.30505 %
$f^{banff}_{p_f}$	3.11707	3.21831	3.14236	-2.3599 %	3.29191	2.2869 %
$f^{banff}_{Shape_{CCoth}}$	3.11707	3.21831	3.22119	0.089515 %	3.21543	-0.089518 %
$f^{banff}_{E_B}$	3.11707	3.21831	3.21749	-0.025439 %	3.21891	0.018569 %
$f^{banff}_{Norm_{CCcoh}}$	3.11707	3.21831	3.22518	0.21355 %	3.21144	-0.21355 %
$f^{banff}_{Norm_{NCcoh}}$	3.11707	3.21831	3.2514	1.0282 %	3.18522	-1.0282 %
$f^{banff}_{Norm_{NCother}}$	3.11707	3.21831	3.22886	0.32779 %	3.20776	-0.32779 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	3.11707	3.21831	3.24258	0.75412 %	3.19404	-0.75412 %
$f^{banff}_{Norm_{NC1\gamma}}$	3.11707	3.21831	3.33932	3.76 %	3.0973	-3.76 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	3.11707	3.21831	3.27411	1.734 %	3.1625	-1.734 %
$f^{banff}_{Norm_{MECBar}}$	3.11707	3.21831	3.2943	2.3612 %	3.14232	-2.3612 %
$f^{SK}_{E;r}$	3.11707	3.21831	3.21831	-1.3799e-14 %	3.21831	-4.1397e-14 %
$f^{SK+FSI}_{6;t,r}RHC$	3.11707	3.21831	3.22805	0.30255 %	3.20857	-0.30255 %
$f^{SK+FSI}_{7;t,r}RHC$	3.11707	3.21831	3.25713	1.2064 %	3.17948	-1.2064 %
$f^{SK+FSI}_{8;t,r}RHC$	3.11707	3.21831	3.24365	0.78746 %	3.19297	-0.78746 %
$f^{SK+FSI}_{9;t,r}RHC$	3.11707	3.21831	3.22378	0.16988 %	3.21284	-0.16988 %
$f^{SK+FSI}_{10;t,r}RHC$	3.11707	3.21831	3.2219	0.11164 %	3.21472	-0.11164 %
$f^{SK+FSI}_{11;t,r}RHC$	3.11707	3.21831	3.22052	0.068625 %	3.2161	-0.068625 %
$f^{SK+FSI}_{12;t,r}RHC$	3.11707	3.21831	3.22172	0.10612 %	3.21489	-0.10612 %
$f^{SK+FSI}_{13;t,r}RHC$	3.11707	3.21831	3.23071	0.38524 %	3.20591	-0.38524 %
$f^{SK+FSI}_{14;t,r}RHC$	3.11707	3.21831	3.23982	0.66827 %	3.1968	-0.66827 %
$f^{SK+FSI}_{15;t,r}RHC$	3.11707	3.21831	3.28217	1.9844 %	3.15444	-1.9844 %
$f^{SK+FSI}_{16;t,r}RHC$	3.11707	3.21831	3.25076	1.0083 %	3.18586	-1.0083 %
$f^{SK+FSI}_{17;t,r}RHC$	3.11707	3.21831	3.2479	0.91957 %	3.18871	-0.91957 %

Table 16: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 4.011×10^{20} POT for antineutrino mode.

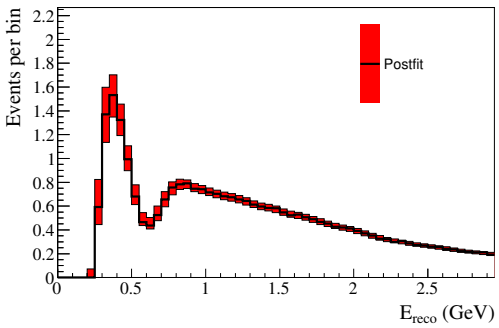
The effects of all the combined systematic uncertainties on the reconstructed energy spectrum of single ring events is shown in Fig. 6 and 7 for the typical oscillation scenario with parameters of Asimov datasets A. In App. D the same plots are shown for different oscillation scenarios. These plots show the total error envelope for the reconstructed energy spectra; this is calculated as the $\pm 1\sigma$ spread with respect to the median of bin contents from 10k toy MC experiments generated with randomized systematic parameters. All systematic parameters are considered and their correlations are taken into account.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 17, 18, 19 and 20 with the typical parameter values given in Tab. 12. For the measurement of δ_{CP} it important to understand the effect of the systematic uncertainties on the ratio of the number of events between neutrino and antineutrino mode samples, which is the consequence of correlations between systematic parameters. If correlations do not reproduce the physics correctly, a fake asymmetry between neutrino and antineutrino oscillations can be seen. On the other hand if correlations exist but are not taken into account, the measurement of δ_{CP} could be biased. Depending on the value of δ_{CP}, correlations or anti-correlations between parameters are more conservative. If true $\delta_{CP} = \pm\pi/2$ the most conservative case is given by anti-correlation in the e-like samples, since the oscillation asymmetry could be hidden by the systematic uncertainties. If true $\delta_{CP} \sim 0$ uncorrelated parameters reduce the sensitivity to δ_{CP}. The effect of the systematic parameters on the ratio between neutrino and antineutrino mode events are shown in Tab. 21. The most critical systematic parameters are σ_{ν_e} and $\sigma_{\bar{\nu}_e}$. In fact they are partially anti-correlated in this analysis and are applied respectively to ν_e and $\bar{\nu}_e$ events.

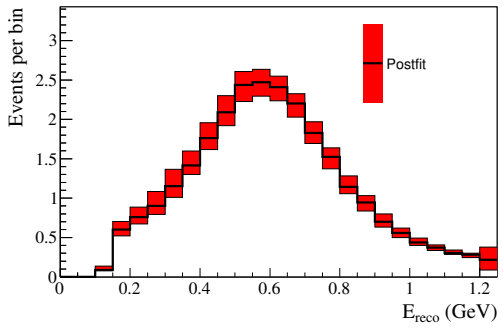
For the systematic effect study, in order to reduce the CPU time, the toys MC were produced using 1-dimensional templates E_{reco} e-like templates, instead of the 2-dimensional E_{reco} Vs θ, used in the analysis. However it has been checked that using 1- or 2- dimensional templates does not make any difference in the drawing of the error envelopes and the effect of the systematic uncertainties on the total number of events.



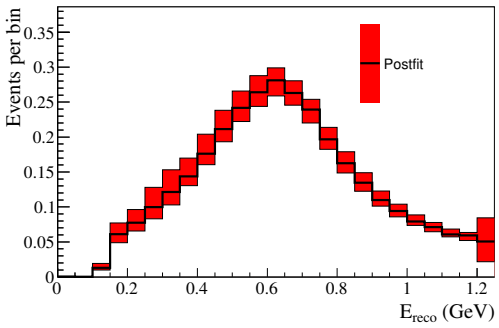
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

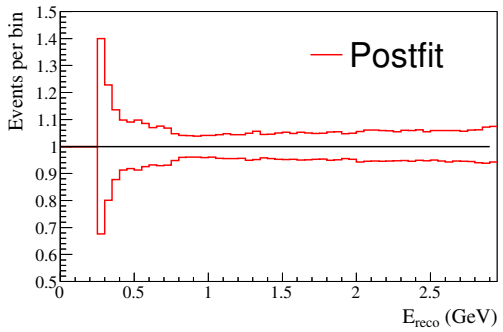


(c) Electron-like, neutrino mode

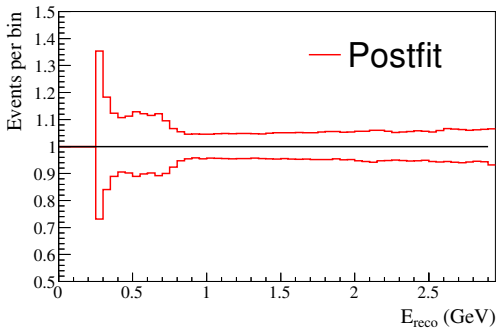


(d) Electron-like, antineutrino mode

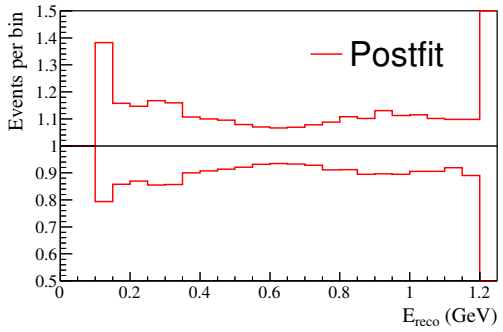
Figure 6: Error envelopes for the energy spectrum as a result of all the systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



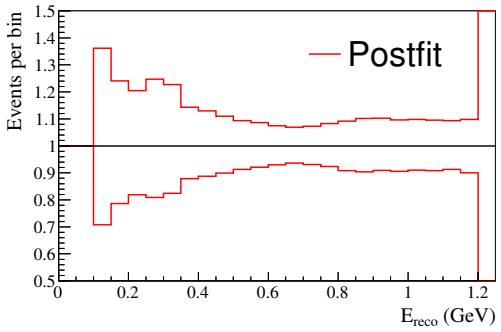
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode



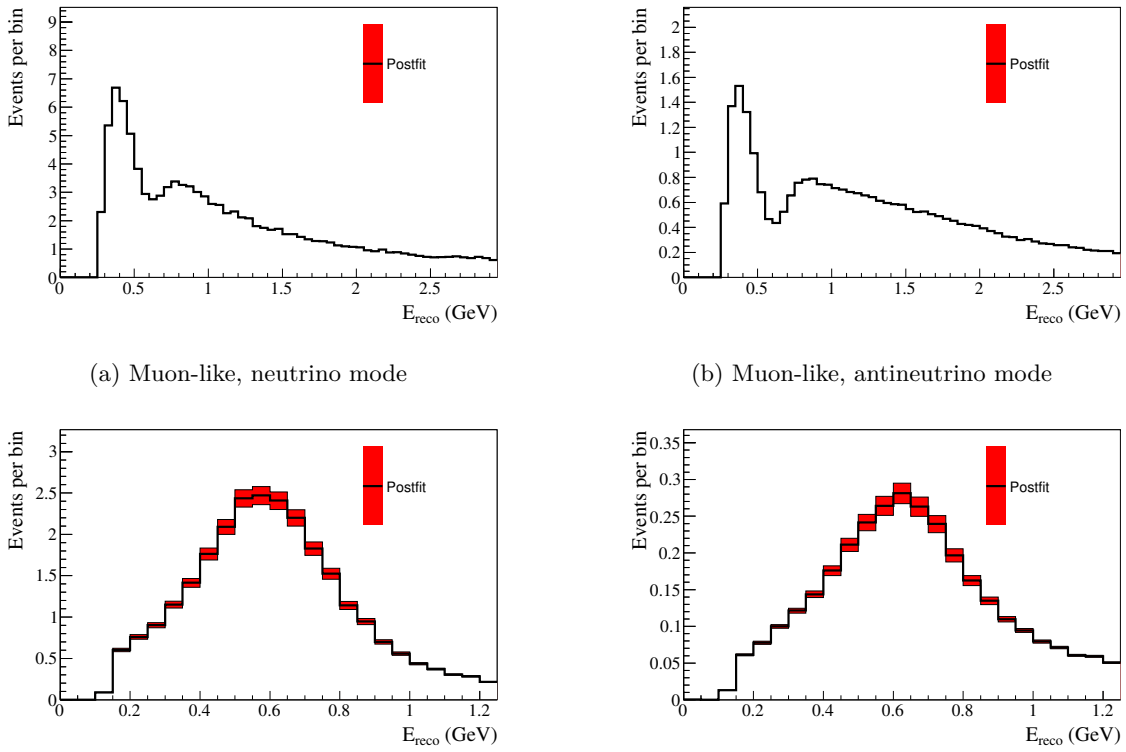
(c) Electron-like, neutrino mode



(d) Electron-like, antineutrino mode

Figure 7: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.

In Fig. 8, 9, 10, 11, 12 the error envelopes of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.



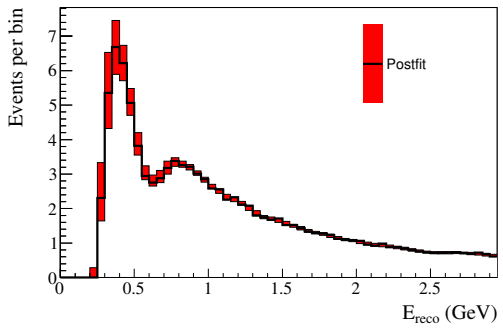
(a) Muon-like, neutrino mode

(b) Muon-like, antineutrino mode

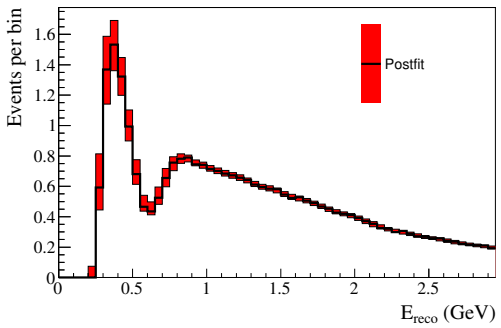
(c) Electron-like, neutrino mode

(d) Electron-like, antineutrino mode

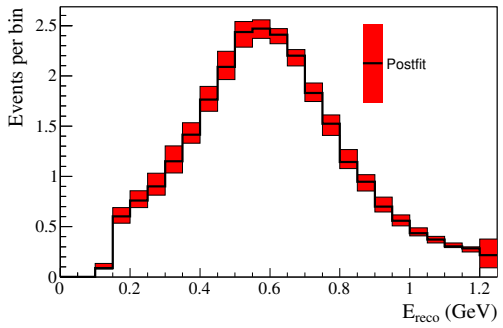
Figure 8: Error envelopes for the energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



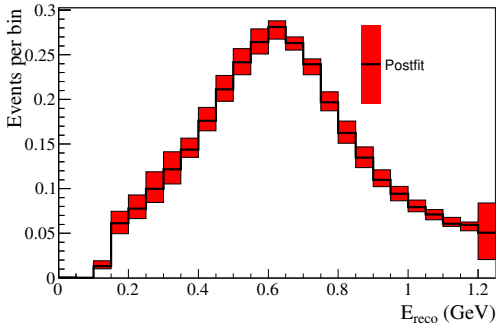
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

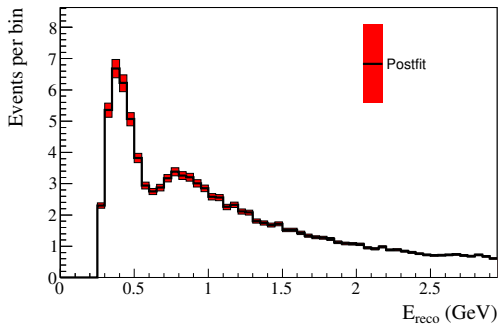


(c) Electron-like, neutrino mode

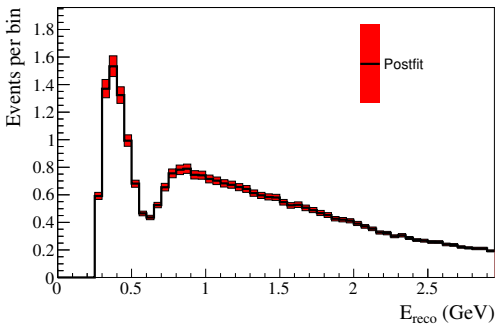


(d) Electron-like, antineutrino mode

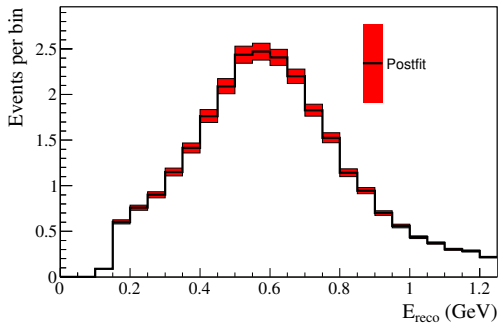
Figure 9: Error envelopes for the energy spectrum as a result of the super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



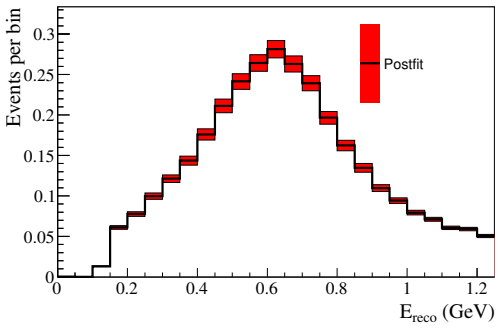
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

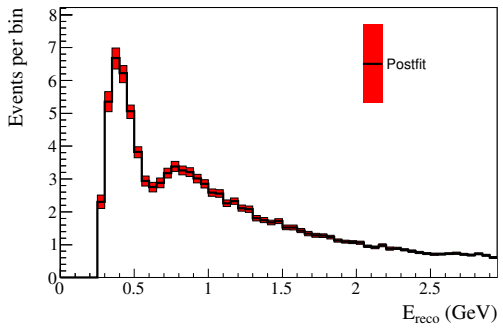


(c) Electron-like, neutrino mode

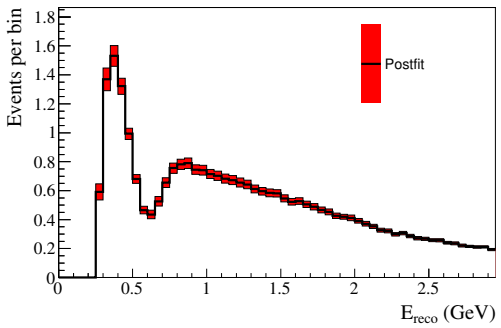


(d) Electron-like, antineutrino mode

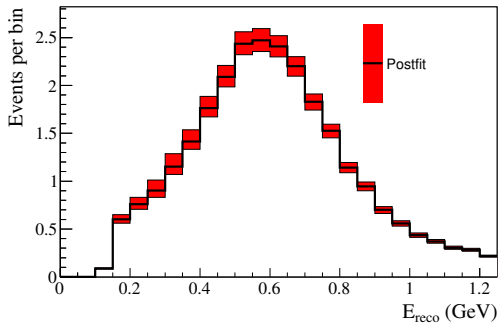
Figure 10: Error envelopes for the energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



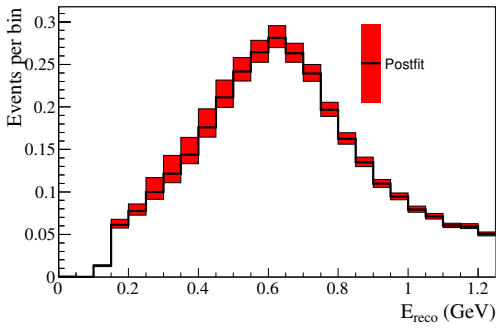
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

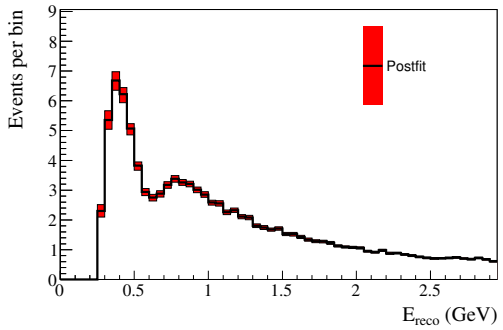


(c) Electron-like, neutrino mode

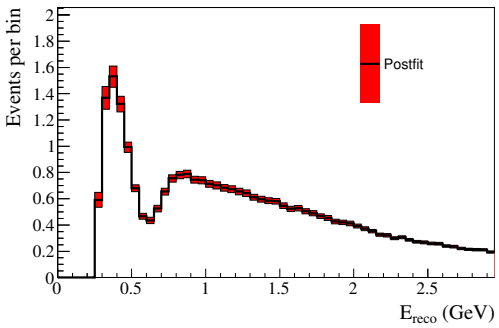


(d) Electron-like, antineutrino mode

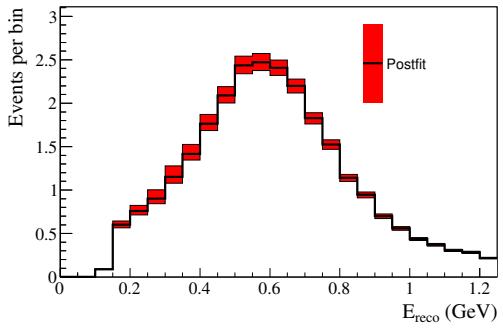
Figure 11: Error envelopes for the energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



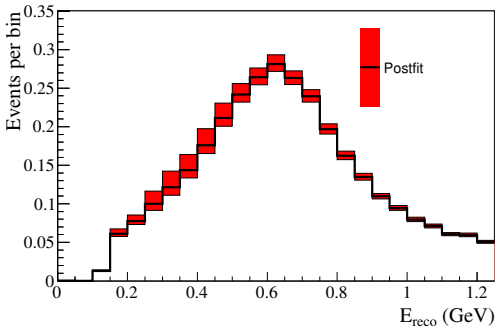
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode



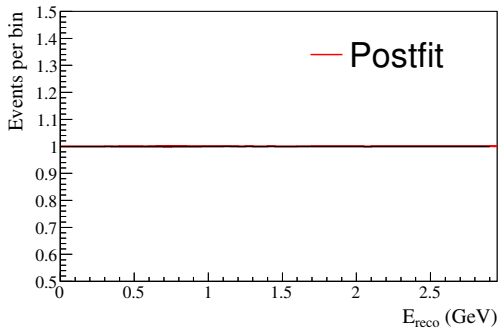
(c) Electron-like, neutrino mode



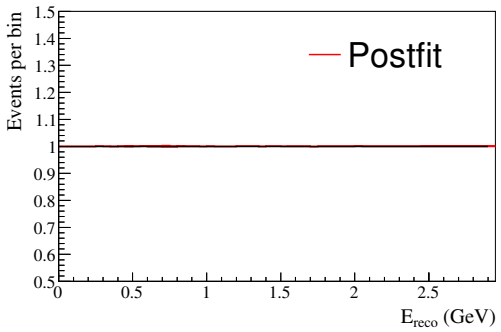
(d) Electron-like, antineutrino mode

Figure 12: Error envelopes for the energy spectrum as a result of the flux plus cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.

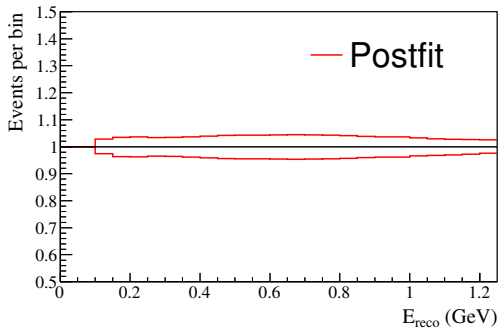
In Fig. 13, 14, 15, 16, 17 the ratio error envelopes of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.



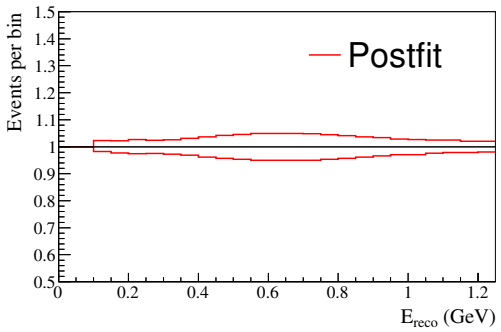
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

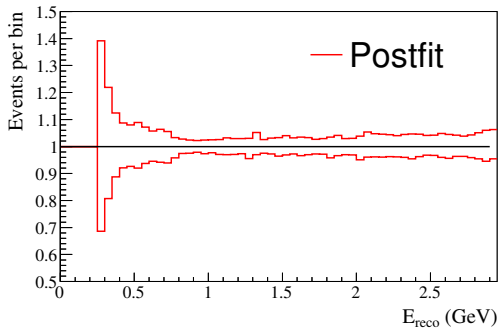


(c) Electron-like, neutrino mode

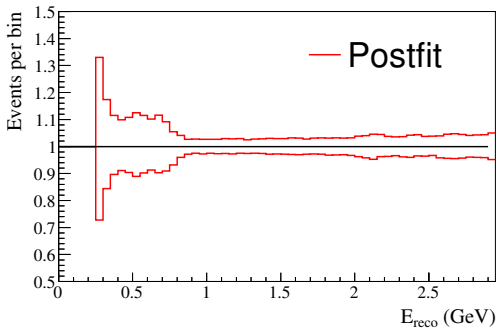


(d) Electron-like, antineutrino mode

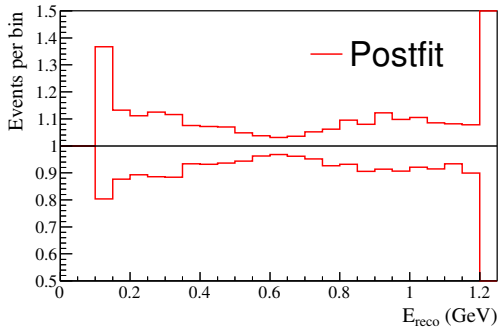
Figure 13: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



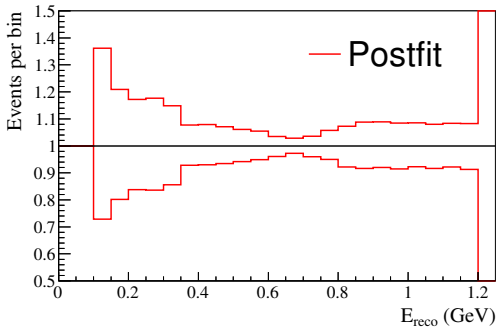
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

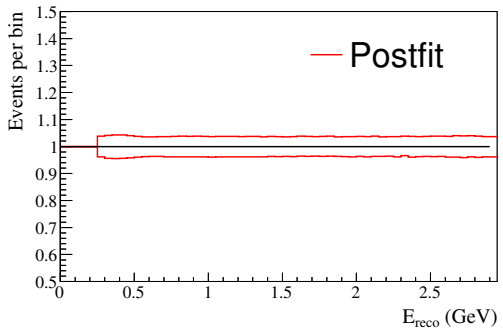


(c) Electron-like, neutrino mode

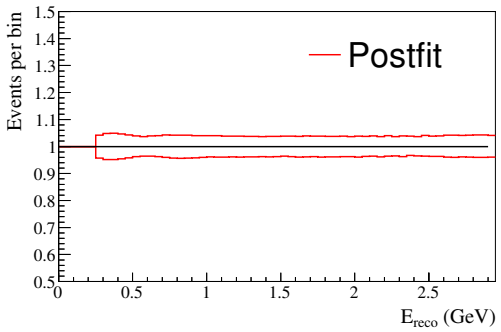


(d) Electron-like, antineutrino mode

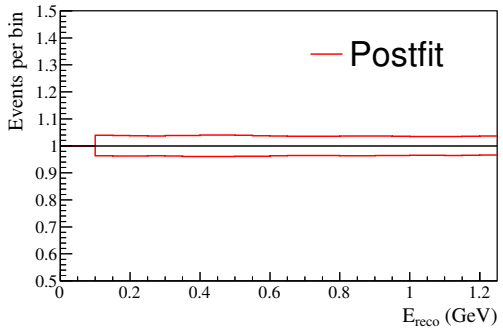
Figure 14: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



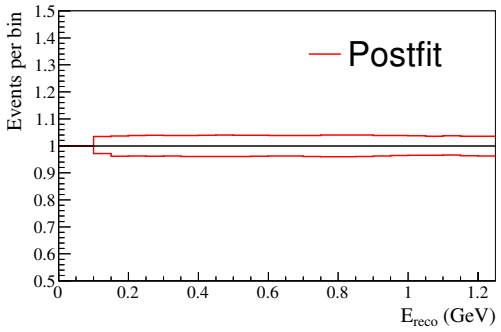
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

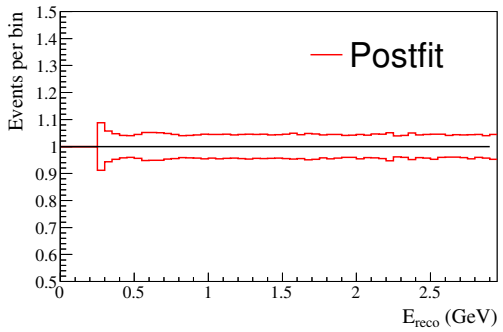


(c) Electron-like, neutrino mode

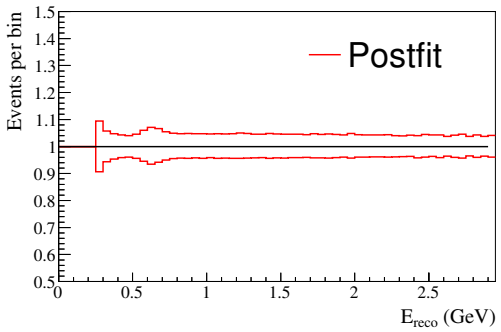


(d) Electron-like, antineutrino mode

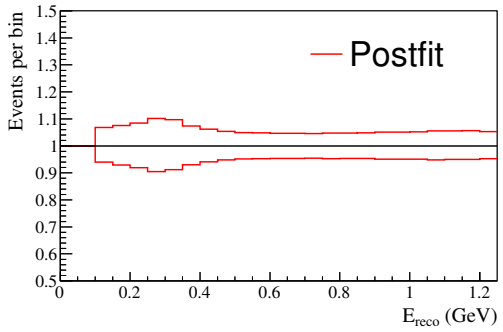
Figure 15: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



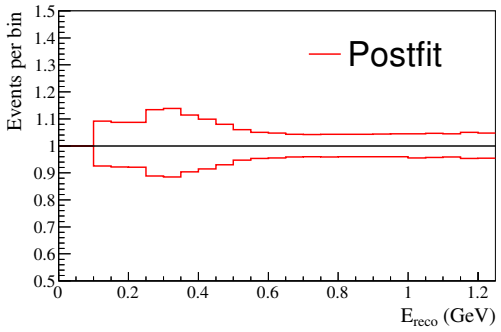
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode

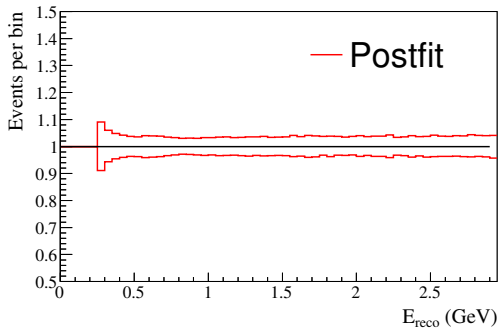


(c) Electron-like, neutrino mode

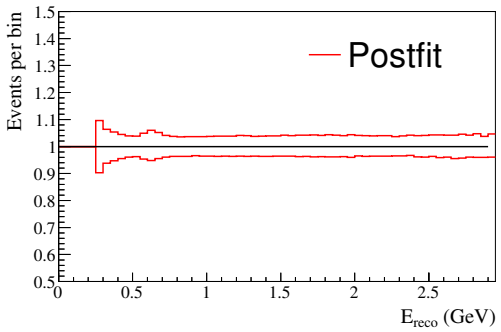


(d) Electron-like, antineutrino mode

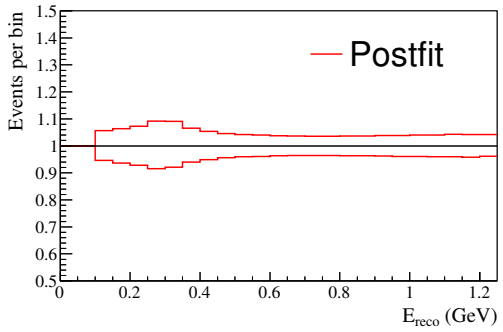
Figure 16: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



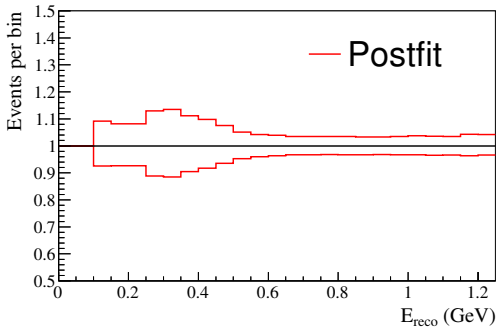
(a) Muon-like, neutrino mode



(b) Muon-like, antineutrino mode



(c) Electron-like, neutrino mode



(d) Electron-like, antineutrino mode

Figure 17: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the flux + cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents is calculated. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	4.15%
Flux	3.59%
XSec Corr Tot	4.02%
MEC (corr)	3.54%
MECbar (corr)	0.20%
NC other (uncorr)	0.77%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.10%
Flux+XSec (ND280 constrained)	2.82%
Flux+XSec	2.92%
Flux+XSec+SKDet+FSI+SI	5.09%
Oscillations	0.03%
All	5.13%
All (pre-fit)	11.9%

Table 17: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.50%
Flux	3.67%
XSec Corr Tot	4.25%
MEC (corr)	3.85%
MECbar (corr)	0.05%
NC other (uncorr)	0.16%
NC 1gamma (uncorr)	1.47%
XSec nue/numu	2.60%
XSec Tot	5.20%
Flux+XSec (ND280 constrained)	2.84%
Flux+XSec	4.13%
Flux+XSec+SKDet+FSI+SI	5.53%
Oscillations	4.16%
All	6.80%
All (pre-fit)	12.6%

Table 18: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.66%
XSec Corr Tot	4.08%
MEC (corr)	2.96%
MECbar (corr)	1.82%
NC other (uncorr)	0.75%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.14%
Flux+XSec (ND280 constrained)	3.27%
Flux+XSec	3.36%
Flux+XSec+SKDet+FSI+SI	5.18%
Oscillations	0.03%
All	5.12%
All (pre-fit)	12.7%

Table 19: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.97%
Flux	3.78%
XSec Corr Tot	4.30%
MEC (corr)	3.00%
MECbar (corr)	2.36%
NC other (uncorr)	0.33%
NC 1gamma (uncorr)	2.93%
XSec nue/numu	1.51%
XSec Tot	5.43%
Flux+XSec (ND280 constrained)	3.19%
Flux+XSec	4.61%
Flux+XSec+SKDet+FSI+SI	6.21%
Oscillations	4.00%
All	7.41%
All (pre-fit)	14.3%

Table 20: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	μ-like RMS/Mean	e-like RMS/Mean
SKDet	0.7%	1.6%
FSI+SI	2.6%	3.5%
SKDet+FSI+SI	2.6%	3.8%
Flux	1.7%	1.7%
XSec NC other (uncorr)	0.0%	0.2%
XSec NC 1γ (uncorr)	0.0%	1.5%
XSec ν_e / ν_μ (uncorr)	0.0%	3.1%
XSec Tot	1.7%	4.2%
Flux+XSec	1.8%	4.1%
Flux+XSec+SDet+FSI+SI	3.2%	5.7%
Oscillations	0.0%	0.8%
All	3.2%	5.8%

Table 21: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

4. Analysis procedure

4.1. Fit method

The goal of this analysis is to measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} and search for CP violation by setting limits on δ_{CP} for a given hypothesis of mass hierarchy, which is fixed to either normal or inverted. All the systematics as well as oscillation parameters, which are nuisance, are marginalised. The treatment of the oscillation parameters is shown in Tab. 12. The parameters of interest will be denoted with the vector $\vec{\theta}$, since more than 1 parameter at a time can be measured. $\vec{\theta}$ can include the combination of all the oscillation parameters listed above.

The *VALOR* analysis uses a binned likelihood-ratio method where the estimation of $\vec{\theta}$ is obtained by comparing the observed and predicted Super-K E_{reco} Vs θ_l spectra for 1-ring events.

Let N be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the Super-K reconstructed energy spectrum prediction (e.g. parameters that control the beam, neutrino interaction simulations, detector response and oscillations).

The log-likelihood function used in the analysis is

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) \tag{3}$$

where n_i^{obs} is the observed number of events in the ith bin and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The log-likelihood function obtained by adding a gaussian penatly term can be written as

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \tag{4}$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematic parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The total likelihood is given by the product of the likelihood terms of each selected sample,

$$\lambda(\vec{\theta};\mathbf{a}) = \prod_{s=1}^{N_{samples}}\lambda_s(\vec{\theta};\mathbf{a}) \tag{5}$$

where $N_{samples}$ is the number of selected samples used in the analysis, or by the sum of $-2\ln\lambda(\vec{\theta},\mathbf{a})$ from all the super-K samples

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s(\vec{\theta},\mathbf{a}) \tag{6}$$

The advantage of the likelihood ratio method over the extended maximum-likelihood method is that, in the large-sample limit, the quantity $-2\ln\lambda(\vec{\theta};\mathbf{a})$ in Eq. 3 has a χ^2 distribution and it can therefore be used as a goodness-of-fit test.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

4.2. Marginalization

In the previous analyses, except [10], a fully frequentist approach, in which all nuisance parameters were profiled, was used. Now a hybrid Bayesian-frequentist method is used. The nuisance parameters are marginalised over while the parameters of interest are minimized with respect to the likelihood distribution. The marginalization of the nuisance parameters give a so-called marginal likelihood that depends on only on a vector of parameters of interest, $\vec{\theta}$.

The marginal likelihood is derived using the following equation:

$$\lambda_{marg}(\vec{\theta}) = \int_A \lambda(\vec{\theta};\mathbf{a})\pi(\mathbf{a})d\mathbf{a} = \frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i) \tag{7}$$

where $\mathbf{a}$ is the set of nuisance parameters, $\pi(\mathbf{a})$ is the prior of the nuisance parameters. n is the number of toys MC produced for the marginalization and $\mathbf{a_i}$ is the single set of nuisance parameters extracted in the "i-th" toy MC. Many sets of $\mathbf{a}$ ($n = 10k$) should be extracted in order to effectively sample the space of the nuisance parameters. The number of extractions depends on the number of nuisance parameters as well as the precision that is required by the measurement.

We derive the marginal log-likelihood, which we will be denoted as χ^2_{marg} from now on since it follows a χ^2 distribution when the Poisson statistic converges to the Gaussian:

$$\chi^2_{marg}(\vec{\theta}) = -2\ln\lambda_{marg}(\vec{\theta}) = -2\ln\left(\frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i)\right) \tag{8}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the n throws, not a weighted average of the log-likelihood. During the marginalization process, random values of the nuisance parameters are drawn according to the prior constraints which come from fits to near detector data and control samples at Super-K. Therefore, each throw is already weighted based on the prior information, and as a result the nuisance parameter values are not included in the calculation of the likelihood. For this reason Eq. 3, which doesn't include the penalty term, is used.

Sometimes the free parameters, which are not marginalised, are constrained with a penalty term. In this case a modified version of Eq. 8 is used:

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + (\vec{\theta} - \vec{\theta_0})^T V^{-1}(\vec{\theta} - \vec{\theta_0}) \tag{9}$$

where $\vec{\theta_0}$ is the vector of nominal values of the parameters of interest, $\vec{\theta}$ is the vector of its fitted values and V is the covariance matrix which relates all the free parameters. If the constrained parameter θ_{constr} is not correlated with the other parameters of interest, V is reduced to the uncertainty $\Delta\theta^{constr}$, and Eq. 9 becomes

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + \frac{(\theta^{constr} - \theta_0^{constr})^2}{(\Delta\theta^{constr})^2} \tag{10}$$

This is the case of $\sin^2\theta_{13}$ when it is measured, i.e. it is parameter of interest, but at the same time it is constrained with the measurement of the reactor experiments. When a penalty term is applied to a parameter of interest it will be specified.

The toys MC used for the marginalization of the nuisance parameters are performed exactly in the same way as described in sec. 3.1, taking into account the correlations, but without statistical fluctuations. First $n = 10k$ distributions of E_{reco} Vs θ_l for e-like and E_{reco} only for μ-like samples are produced using the extracted systematic parameters for 1×10^{20} POT so that they may be scaled appropriately when comparing to datasets of differing POT. Finally they are stored in output files which can be used for whatever needed fit. A fit on a single dataset is performed as following:

- all the n toy MC distributions are read from the files,
- each distribution is normalised to the number of POT of the data set,
- for each toy distribution Eq. 3 is calculated and $\lambda(\vec{\theta},\mathbf{a}) = e^{-(-2\ln\lambda(\vec{\theta},\mathbf{a}))/2}$ is obtained,
- the χ^2 value is finally obtained, using Eq. 8, for a given set of parameter values of interest, $\vec{\theta}$. If $\vec{\theta}$ is constrained, eq. (10) is used.

This procedure is followed for each point in the $\vec{\theta}$ grid, where the binning is so fine that it does not impact on the precision of the measurement. Finally the best fit is found by looking for the $\vec{\theta}$ grid point with the minimum χ^2_{bf} value. This point corresponds to the best-fit parameters $\vec{\theta}_{bf}$. The minimization can be also performed by using a discrete descent gradient algorithm implemented in the VALOR framework that allows the minimum χ^2 to be found without the need of calculating the full χ^2 distribution across all grid points.

This method is employed when a large number of fits is required, such as the calculation of critical values for Feldman-Cousins contours, as it drastically reduces the amount of CPU time required. A validation of the minimization algorithm will be shown in sec. 5 for the analyses where it is used.

4.3. Best-fit spectra with marginalization

When profiling, the best fit spectra are defined by the MC prediction that gives the largest likelihood values.

However, since the nuisance parameters are marginalised, there does not exist a single set of parameters which corresponds to the best fit. Therefore a new method has been developed to represent the best fit spectra which is consistent with the oscillation results.

The following procedure is used. From each toy MC i, used to marginalise the nuisance parameters, a different spectrum, with $N^j(\vec{\theta}_{bf};\mathbf{a})$ number of events in each bin j, is produced as well as a likelihood $\lambda(\vec{\theta}_{bf},\mathbf{a_i})$ value, calculated using Eq. 7, where $\vec{\theta}_{bf}$ is the best-fit value of the free parameters vector and $\mathbf{a_i}$ is the vector of nuisance parameters corresponding to the toy i. The best-fit number of events in the bin j of the spectrum is calculated as

$$N^j_{marg} = \frac{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf};\vec{f}_i)\cdot N^j(\vec{\theta}_{bf};\vec{f}_i)}{\sum_{i=1}^{n}\lambda(\vec{\theta}_{bf};\vec{f}_i)} \tag{11}$$

where n is the number of toys MC.

One can easily find a relation between the profiled and marginalised best-fit spectra. In the limit of infinite number of toys MC, i.e. the full nuisance parameters space is scanned, the likelihood can be written as

$$\lambda_{marg}(\vec{\theta}_{bf}) = \frac{1}{n}\sum_{i=1,\ i\neq bf}^{n}\Big[\lambda(\vec{\theta}_{bf};\mathbf{a_i}) + \lambda_{\mathbf{prof}}\Big] \tag{12}$$

where $\lambda_{prof} = \lambda(\vec{\theta}_{bf};\mathbf{a_{bf}})$ is the profiled best-fit likelihood. If we denote as $N^j_{prof} = N^j_i(\vec{\theta}_{bf};\mathbf{a_{bf}})$ the number of events in the bin j obtained with profiling, Eq. 11 can be written as

$$N^j_{marg} = \frac{\sum_{i=1,\ i\neq bf}^{n}\lambda(\vec{\theta}_{bf};\mathbf{a_i})\cdot\mathbf{N^j}(\tilde{\theta}_{\mathbf{bf}};\mathbf{a_i}) + \lambda_{\mathbf{prof}}\cdot\mathbf{N^j_{prof}}}{\sum_{i=1}^{n}\lambda(\vec{\theta}_{bf};\mathbf{a_i})} \tag{13}$$

4.4. Confidence intervals

After the fit is completed, the full χ^2 distribution is then shifted with respect to $\chi^2(\vec{\theta}_{bf})$:

$$\Delta\chi^2(\vec{\theta}) = \chi^2(\vec{\theta}) - \chi^2(\vec{\theta}_{bf}) \tag{14}$$

Confidence intervals are set on $\Delta\chi^2(\vec{\theta})$. If

$$\Delta\chi^2(\vec{\theta}) > \Delta\chi^2_{crit} \tag{15}$$

where $\Delta\chi^2_{crit}$ is the critical value for a certain confidence level X%(CL), that value of $\vec{\theta}$ is excluded at the X% CL. In the gaussian approximation constant values of $\Delta\chi^2_{crit}$ can be used. It will be called "constant $\Delta\chi^2$ method".

However in some cases the gaussian approximation is not reliable and the "Feldman-Cousins method" [54] must be used. In this method a different critical value for each point of the $\vec{\theta}$ grid $\Delta\chi^2(\vec{\theta})_{crit}$ is calculated. The procedure consists of the following steps:

1. many toys MC (e.g. 10k) are produced assuming the oscillation hypothesis of the grid point $\vec{\theta}$ and taking into account both the statistical and systematic uncertainties;
2. for each toy first $\chi^2(\vec{\theta}_{fix})$ is obtained by minimizing $\chi^2(\vec{\theta})$ where the oscillation parameters $\vec{\theta}$ are kept fixed to the values corresponding to the grid point and the nuisance parameters are marginalised. Then $\chi^2(\vec{\theta})$ is minimized, leaving the oscillation parameters $\vec{\theta}$ as free and $\chi^2(\vec{\theta}_{bf})$ is obtained.

3. The steps 2 and 3 are repeated for each toy MC and a distribution of $\Delta\chi^2(\vec{\theta})$ ($f(\Delta\chi^2)$) is obtained.

4. The critical value for $X\%$ CL, $\Delta\chi^2(\vec{\theta})_{crit}$, can be defined as:

$$\Delta\chi^2_{crit} : \int_{-\infty}^{\Delta\chi^2_{crit}} f(\Delta\chi^2) d(\Delta\chi^2) = X\% \tag{16}$$

5. This procedure is repeated for all the points of the $\vec{\theta}$ grid and several critical values are computed

Once a critical value, $\Delta\chi^2(\vec{\theta})_{crit}$, is computed for each oscillation hypothesis in the $\vec{\theta}$ grid, the confidence intervals are set by the condition of Eq. 15.

Several analyses are performed and confidence intervals are computed for:

- 1D δ_{CP} with reactor constraint: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 1D δ_{CP} T2K only: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the uniform prior, shown in Tab. 22, is used.
- 1D $\sin^2\theta_{13}$ T2K only: the parameter of interest is $\sin^2\theta_{13}$ and all the other oscillation parameters are marginalised;
- 1D Δm^2_{32} (Δm^2_{13}) with reactor constraint: the parameter of interest is Δm^2_{32} (Δm^2_{13}) and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 1D $\sin^2\theta_{23}$ with reactor constraint: the parameter of interest is $\sin^2\theta_{23}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 2D $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is a free parameter, the gaussian prior of reactor measurements shown in Tab. 22, is used, through a penalty term as in Eq. 10.
- 2D $\sin^2\theta_{13}$ Vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ Vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{23}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D Δm^2_{32} Vs δ_{CP} T2K only: the parameters of interest are Δm^2_{32} and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ Vs Δm^2_{32} with reactor constraint: the parameters of interest are $\sin^2\theta_{23}$ and Δm^2_{32} (Δm^2_{13}) while all the other oscillation parameters are marginalised;

Each confidence interval is calculated for both normal and inverted hierarchy, which is then profiled.
The 1D χ^2 confidence intervals with "constant $\Delta\chi^2$ method" are used to set the $\pm1\sigma$ ranges.
Each point of the $\vec{\theta}$ grid corresponds to the center of the histogram bins, i.e. the χ^2 values are calculated at each histogram bin center. The ranges and binnings used for the analyses listed above are shown in Tab. 22. All the 1-dimensional analyses, where the precision on the best-fit point is important and must be quoted, are performed with a fine binning. Since 2-dimensional analyses are performed only to set contours with "constant $\Delta\chi^2$ method" and a fit with a very high precision is not needed to find the best-fit point, courser binning are used, without affecting the contour. Furthermore a very fine 2-dimensional binning would require much more CPU time and storage space. In the future $\bar{\nu}/\nu$ joint analysis releases when the "Feldman-Cousins method" will be needed also for 2-dimensional fits, a finer binning will be used.

Parameter	Min	Max	Step	Bins
	1-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/200$	201
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 200	201
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 200	201
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	3×10^{-3}	$10^{-3}/200$	201
	2-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/50$	51
$\sin^2\theta_{13}$ (reactors)	0.015	0.036	0.021 / 81	81
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 81	81
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 81	81
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	3×10^{-3}	$10^{-3}/21$	21

Table 22: Grid limits and spacing used for all the 1-dimensional and 2-dimensional marginalization fits and confidence regions. The minimum and maximum values correspond to the center of the χ^2 histogram first and last bin.

4.5. Contours with profiled mass hierarchy

As already pointed out, the constant $\Delta\chi^2$ method gives the correct coverage when the gaussian approximation is valid and is very useful because it does not require a large amount of CPU time. However, we know that in some cases the constant $\Delta\chi^2$ method fails to give the correct coverage and can provide only approximate confidence intervals. An example of this is the case when the constraint on a parameter is not precise enough to keep it away from a physical boundary. For cases such as these a toy MC study is required in order to estimate the correct critical values (see section 4.4).

Also, when considering discrete parameters, such as the mass hierarchy, the $\Delta\chi^2$ is not expected to follow a χ^2 distribution but more likely to follow a Bernoulli distribution as found in [59]. This is an important distinction as the exclusion significance one can interpret from a constant $\Delta\chi^2$ contour is heavily dependant on the distribution the $\Delta\chi^2$ is expected to take.

This means that when the constant $\Delta\chi^2$ method is used in order to set limits as a function of some oscillation parameters as well as the mass hierarchy, the constant critical values we usually use do not give the right coverage for the disfavored mass hierarchy hypothesis contour.

For this reason the data fit 2-dimensional confidence intervals, which are calculated for both the hierarchy hypotheses, will be computed in two different ways:

- normal and inverted hierarchy contours are obtained independently with respect to their own minimum ("independent")
- the global minimum between the normal and inverted hierarchy hypotheses is used to set both the contours ("global");

These different ways to show the confidence intervals correspond to the two extremes, where the "independent" and "global" contours provide respectively the most and the least conservative results.

Since for the 1-dimensional fits the $\Delta\chi^2$ distribution are shown, there is no need to provide results using both "independent" and "global" approaches. However the reader should remember that the constant critical values shown in the figures hold only for the $\Delta\chi^2$ distribution corresponding to the best-fit mass hierarchy hypothesis.

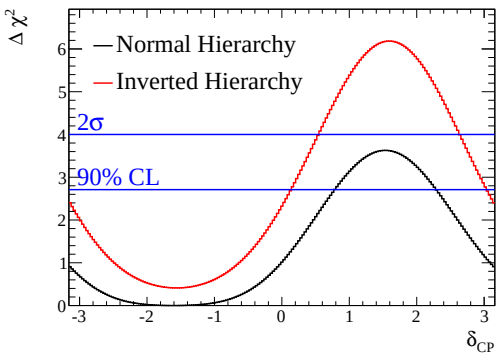
5. Validation of the $\bar{\nu}/\nu$ joint analysis

The validation of the fit is performed first by comparing the confidence intervals, obtained with Asimov data sets corresponding to the oscillation parameters in tab. 12 with the other group analyses [42]. Then the coverage study is performed for 1-dimensional δ_{CP} with reactor constraint fits, in order to verify if the "constant $\Delta\chi^2$" method is reliable or the "Feldman-Cousins" method is needed.

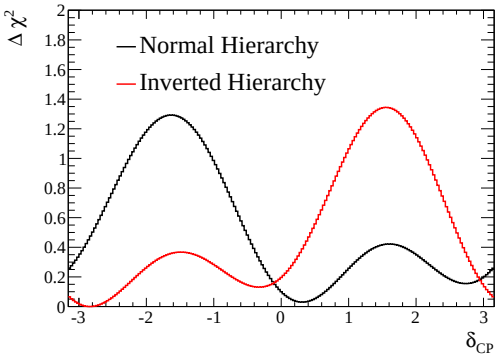
5.1. Sensitivity studies

The critical values of constant $\Delta\chi^2$ method in 1-dimensional confidence intervals are given by $Z\sigma = \sqrt{\Delta\chi^2}$ where Z is the number of standard deviations that defines the confidence level. For 90% CL the critical value is 2.71. The critical values for 2-dimensional fits are 2.3 and 4.61 respectively for 68% and 90% CL.

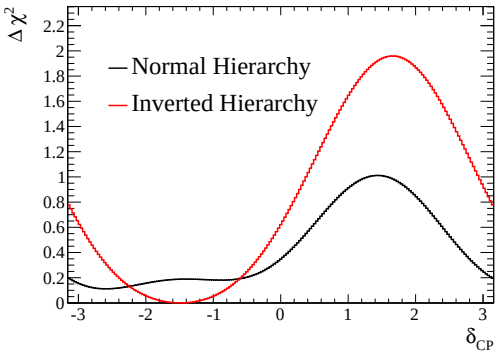
In Fig. 18 the sensitivity to δ_{CP} with reactor constraint and T2K only is shown for the Asimov data sets A and B.



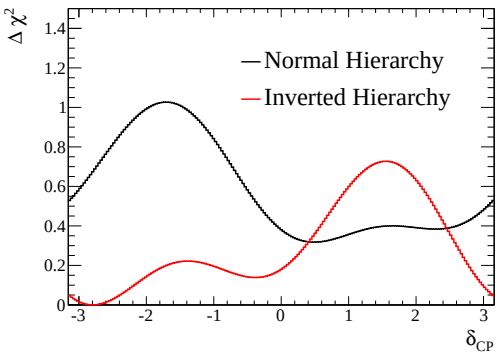
(a) Asimov A: with reactor constraint



(b) Asimov B: with reactor constraint



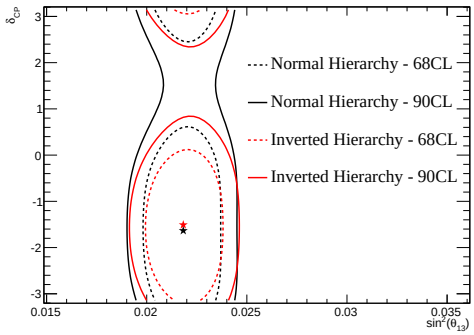
(c) Asimov A: with T2K only



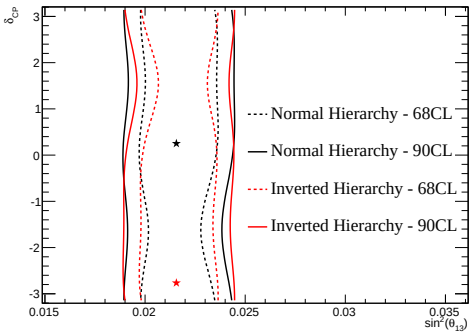
(d) Asimov B: with T2K only

Figure 18: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} for the Asimov data sets A and B of Tab. 12 corresponding to the Run 1-6 statistics is compared to the result obtained by the other analysis groups. The mass hierarchy is minimized: both the normal and inverted hierarchy χ^2 distributions are shifted to the same global best-fit χ^2 value, which is taken as the minimum between normal and inverted hierarchy.

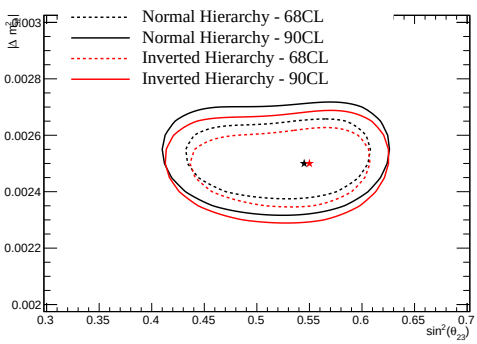
Also the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint and T2K only data, and $\sin^2\theta_{23}$ Vs m^2_{32} (m^2_{13}) with reactor constraint are shown in Fig. 19 and 20, where normal and inverted hierarchy contours are set independently, i.e. each calculated with respect to its own minimum χ^2 value and not the global minimum as a function of the mass hierarchy.



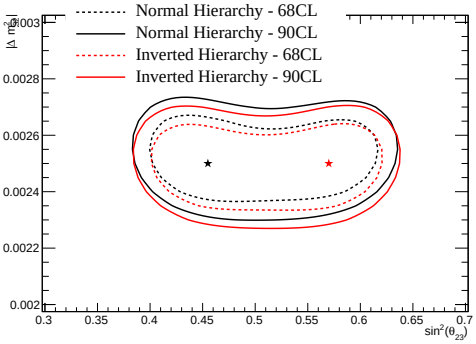
(a) Asimov A: $\sin^2\theta_{13}$ Vs δ_{CP}



(b) Asimov B: $\sin^2\theta_{13}$ Vs δ_{CP}

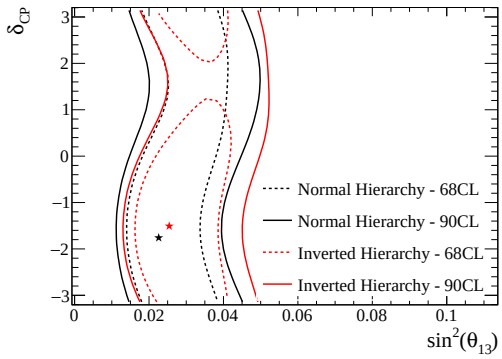


(c) Asimov A: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13})

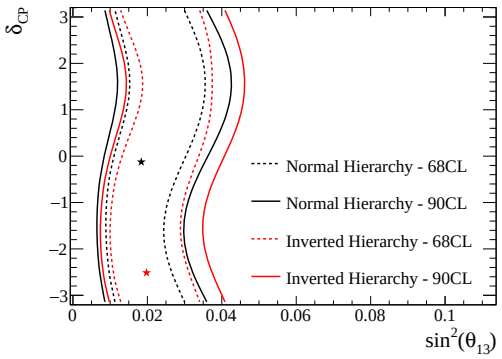


(d) Asimov B: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13})

Figure 19: The contours at 68% and 90% CL with reactor constraint for the Asimov data sets A and B of tab. 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.



(a) Asimov A with reactor constraint



(b) Asimov B with reactor constraint

Figure 20: The contours at 68% and 90% CL for $\sin^2\theta_{13}$ Vs δ_{CP} for the Asimov data sets A and B of tab. 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.

5.2. Coverage study of δ_{CP}

The fit validation is performed also by making coverage studies for the fit of the mass hierarchy (MH) and δ_{CP} with the reactor constraint, which is the main result of this release of the $\bar{\nu}/\nu$ joint fit, in order to understand if the "Feldman-Cousins method" is needed. The coverage is defined as the number of times the true values of δ_{CP}^{true} and MH^{true}, used to generate the toys MC, is contained in the confidence interval defined by eq. 15:

$$\Delta\chi^2(\delta_{CP}, \mathrm{MH}) = \chi^2(\delta_{CP}^{true}, \mathrm{MH}^{true}) - \chi^2(\delta_{CP}^{bf}, \mathrm{MC}^{bf}) < \Delta\chi^2_{crit} \tag{17}$$

where δ_{CP}^{bf} and MH^{bf} are the best-fit values. Adopting the "constant $\Delta\chi^2$" method, the critical value $\Delta\chi^2_{crit} = \Delta\chi^2_{crit}$ at a certain $X\%$ confidence level is used for any value of δ_{CP} and MH. Many toys MC are performed (10k) and for each one eq. 17 is tested. The fit has a good coverage if the number of toys MC, for which eq. 17 is satisfied (N^X_{toys}), is equal or close to $X\%$ within the statistical uncertainty of the number of toys MC. If the $N^X_{toys} > X\%$, then the fit has over-coverage, that means the confidence intervals should be tighter than what obtained with the "constant $\Delta\chi^2$ method", while if $N^X_{toys} < X\%$, then the fit has under-coverage and excludes a wider range of values than in the correct case. In Fig. 21 the coverage study of δ_{CP} 1-dimensional fit with reactor constraint is shown for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$ and both mass hierarchies.



(a) $\delta_{CP}^{true} = XXX$ with reactor constraint



(b) $\delta_{CP}^{true} = XXX$ with reactor constraint



(c) Critical values done with the "Feldman-Cousins" method as a function of δ_{CP}^{true}, compared with constant $\delta\chi^2$ critical values



(d) Coverage as a function of δ_{CP}^{true}

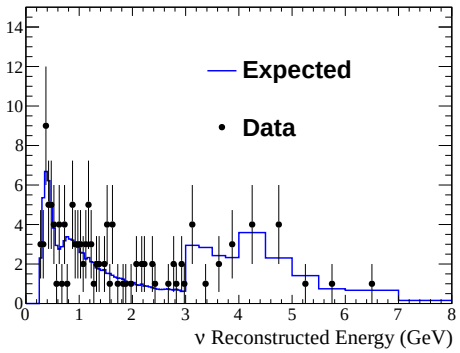
Figure 21: Coverage study of δ_{CP} with reactor constraint for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$ assuming normal hierarchy. All the other oscillation parameters are marginalised and the mass hierarchy is kept fixed to normal in the fit. The statistics corresponds to the Run 1-6 data set.

TODO: ADD COMMENT ON COVERAGE STUDY

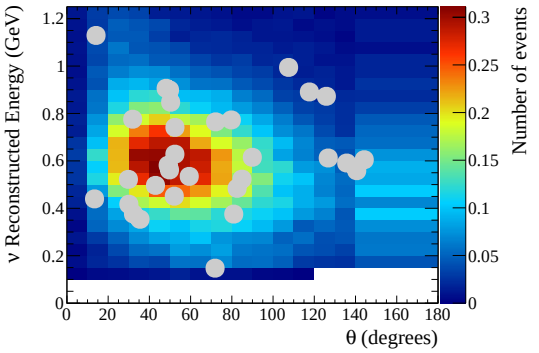
6. Results of the $\bar{\nu}/\nu$ joint analysis with the Run 1-6 dataset

The results of the Run 1-6 data set fit for an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode is shown in this chapter. In fig. 22 the predicted spectra of μ-like and e-like samples are compared to the observed data. In table 23 the expected and observed number of events in Run 1-6 data set are shown. The predicted number of events are shown for the oscillation parameters in table 12 of Asimov data sets A and B. All the predicted and observed events are in good agreement. A small excess of events with respect to the expectation is observed in the neutrino mode e-like samples, though it is still within 1σ of the statistical uncertainty. This trend is mostly inherited from the Run 1-4 data set, since only 2 neutrino mode e-like events are observed in Run 5+6. Furthermore the latest global fit of the reactor experiments gives a lower value of $\sin^2\theta_{13}$ with a much smaller uncertainty (see Sec. 3.1 and Tab. 12). It is interesting to see that the number of events of antineutrino mode e-like events is in perfect agreement with the assumption of $\delta_{CP} = -1.601$, which is the best-fit value of the Run 1-4 data set. This means that, for what concerns the event rate, we expect to see agreement between neutrino and antineutrino mode data in the fit of δ_{CP}, though the statistics are not expected to be enough to reject $\delta_{CP} = 0$ at any significant confidence level.

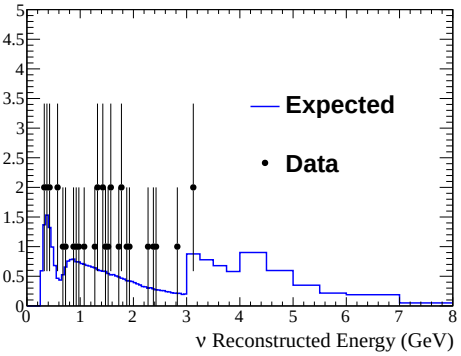
The comparison of the confidence intervals obtained by the fit of the Run 1-6 data set among all the oscillation analysis groups is shown in [42].



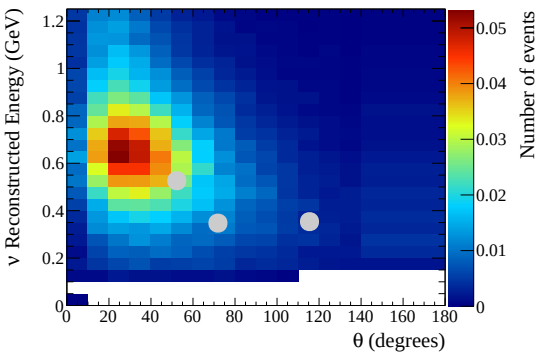
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode

Figure 22: The predicted spectra compared to the Run 1-6 dataset (dots), which corresponds to an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode, are shown. The expected distribution corresponds to the oscillation parameters of the Asimov data set A shown in table 12. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

Beam mode	Sample	Exp. Asimov A	Exp. Asimov B	Exp. Not Osc	Observed	Best fit
neutrino	μ-like	125.914	130.464	484.737	124	125.688
neutrino	e-like	26.594	19.954	5.698	31	26.969
antineutrino	μ-like	34.284	35.032	99.02	34	34.114
antineutrino	e-like	3.218	3.355	1.253	3	3.344

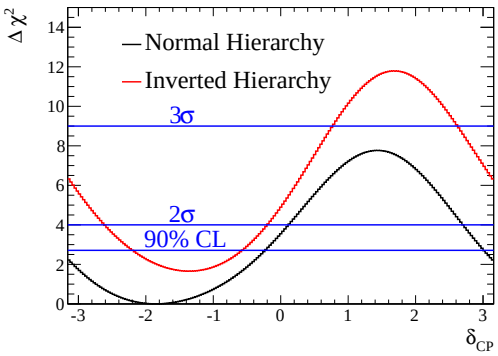
Table 23: The observed, expected and best-fit number of events in Run 1-6 data set for an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning and the oscillation parameters shown in Tab. 12. The total number of events of the best-fit spectra is obtained by using Eq. 11.

6.1. 1-dimensional χ^2 distributions and best-fit

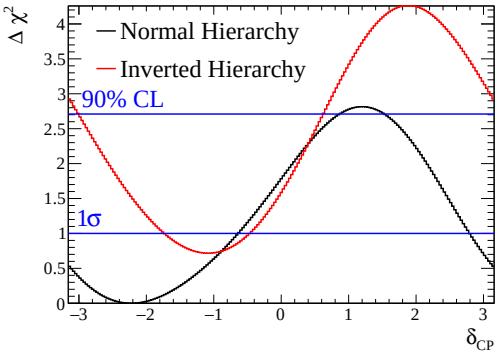
The same 1-dimensional χ^2 distributions as in sec. 5.1 are produced and shown in Fig. 23. The best-fit value of δ_{CP} when the reactor constraint is applied is -1.854 not as close to $-\pi/2$ as the Run 1-4 fit result, which best-fit value was -1.601 (see Tab. 12). However the difference of χ^2 between δ_{CP}^{bf} (best-fit value) and $\delta_{CP} = -\pi/2$ is smaller than 0.1. So the result is consistent with $\delta_{CP} = -\pi/2$ as well as the Run 1-4 data fit.

Looking at the χ^2 distribution of $\sin^2\theta_{13}$ with T2K only data we can see that the reactor measurements are in agreement with the T2K data at about 1σ level if $\delta_{CP} \sim -\pi/2$ and normal hierarchy is assumed.

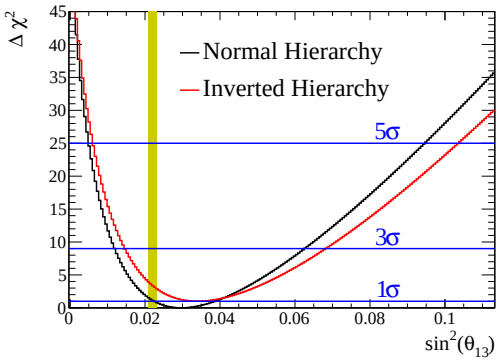
In Fig. 24 the χ^2 distributions as a function of δ_{CP} with reactor constraint is obtained by running the analysis using either only neutrino or antineutrino mode samples for the Run 1-6 data set, in order to check the compatibility of neutrino and antineutrino mode data. The result is mainly driven by the excess of the observed number of events in e-like neutrino mode sample compared to what is expected. Though the



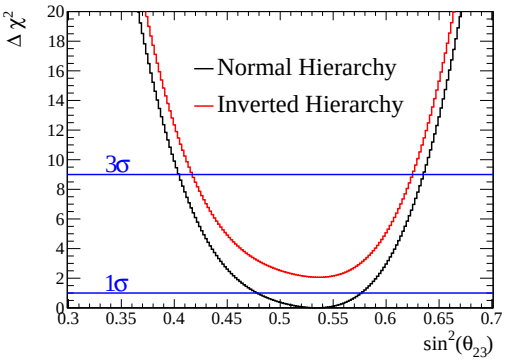
(a) δ_{CP} with reactor constraint



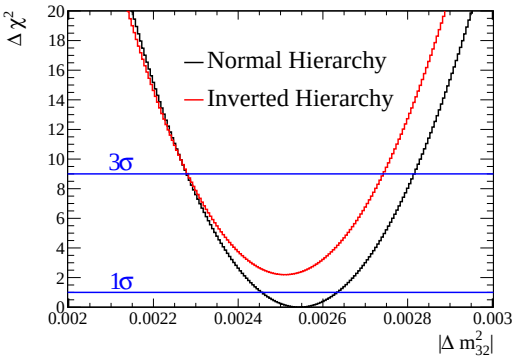
(b) δ_{CP} with T2K data only



(c) $\sin^2\theta_{13}$ with T2K data only



(d) $\sin^2\theta_{23}$ with reactor constraint



(e) Δm^2_{32} (Δm^2_{13}) with reactor constraint

Figure 23: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.

significance given by the antineutrino mode samples is still low, also given by the fact that $\sin^2\theta_{23}$ is not constrained by the antineutrino mode μ-like sample as much as by the neutrino mode μ-like sample, it is interesting to note that the antineutrino data also prefer a value of δ_{CP} close to $-\pi/2$.

In fig. 25 the χ^2 distributions as a function of δ_{CP} is obtained by performing a rate-only analysis; instead of using the shape information of the E_{reco} Vs θ_l spectra, the total number of events of each sample is fitted, reducing the E_{reco} Vs θ_l template to 1 bin. Then the best-fit value of δ_{CP} is -1.602 closer to $-\pi/2$ with respect to the case where the shape information is used.

In Fig. 26 the confidence intervals obtained with the "Feldman-Cousins" method are shown for δ_{CP} with

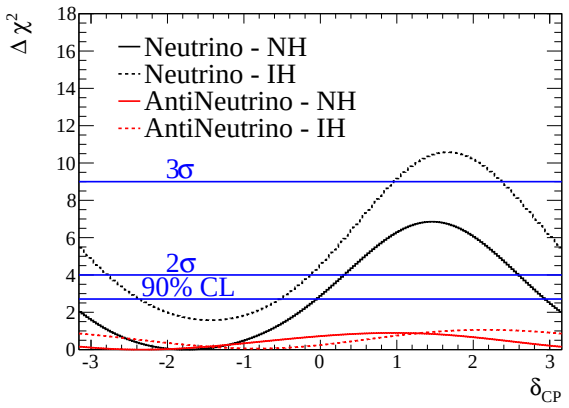


Figure 24: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set as a function of δ_{CP} with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22.

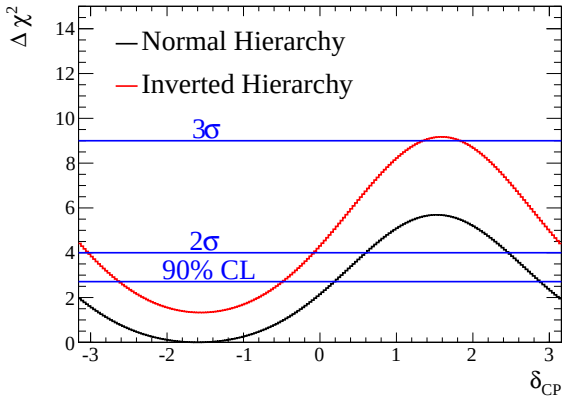


Figure 25: Measured $\Delta\chi^2$ distributions with a rate-only analysis using the full run 1-6 data set as a function of δ_{CP} with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22.

reactor constraint. The critical values are calculated as a function of δ_{CP} as well as the mass hierarchy, as shown in section 4.4.

In Tab. 24 the best-fit values as well as the 1σ ranges for each oscillation parameter are shown, for both normal and inverted hierarchy. The 1σ ranges are extracted for each oscillation parameter from the 1-dimensional $\Delta\chi^2$ distributions shown in Fig. 23, where one oscillation parameter is free and all the other oscillation as well as the systematic parameters are marginalised. The mass hierarchy is minimized: for both normal and inverted hierarchy the 1σ ranges are calculated with respect to the same global minimum. Since this approach is not entirely correct and there does not exist any procedure to set more reliable confidence intervals which does not consist of toy MC studies (see section 4.5), the advice to the reader is to give more importance to the difference in χ^2_{marg} between normal and inverted hierarchies in the respective best-fit points, instead of the 1σ ranges of inverted hierarchy, which are smaller by construction. It is also worth to notice that the 1σ ranges for normal hierarchy (best-fit) are set correctly with the constant $\Delta\chi^2$ method in the gaussian approximation.

The best-fit spectra, shown in fig. 27 are computed as described in sec. 4.3 and are compared to the unoscillated nominal spectra. For e-like samples, where 2-dimensional templates are used, the projection onto the reconstructed neutrino energy axis is shown. All the systematic and oscillation parameters are marginalised following the procedure described in sec. 4.3, while $\sin^2\theta_{23}$, δ_{CP} and the mass hierarchy were

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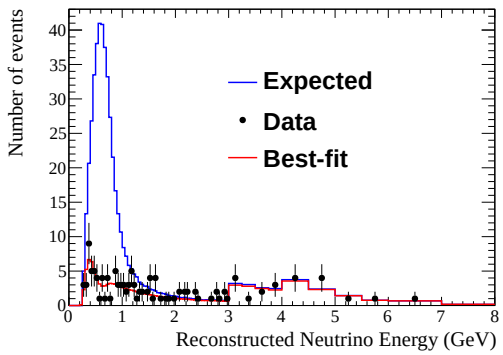
Figure 26: MEASURED FELDMAN-COUSINS DELTACP W/ REACTORS Measured $\Delta\chi^2$ distributions with the full run 1-6 data set as a function of δ_{CP} with reactor constraint, δ_{CP} with T2K data only, $\sin^2\theta_{13}$ with T2K data only, $\sin^2\theta_{23}$ with reactor constraint and Δm^2_{32} (Δm^2_{13}) with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.

	Reactors	**Normal Hierarchy**			**Inverted Hierarchy**		
Parameter		Best-fit	χ^2_{min}	$\pm1\sigma$ interval	Best-fit	χ^2_{min}	$\pm1\sigma$ interval
δ_{CP}	YES	-1.854	296.51	[-2.737; -0.859]	-1.351	298.16	-
δ_{CP}	NO	-2.231	299.49	[-π; -0.640] & [2.793;+π]	1.100	300.20	[-1.735;-0.469]
$\sin^2\theta_{13}$	NO	0.0299	297.09	[0.0224; 0.0387]	0.0339	298.09	-
$\sin^2\theta_{23}$	YES	0.534	295.92	[0.478; 0.577]	0.536	297.98	-
Δm^2_{32} or Δm^2_{13}	YES	2.545	295.36	[2.455; 2.634]	2.510	297.56	-
(10^{-3} eV$^2/c^4$)	YES						

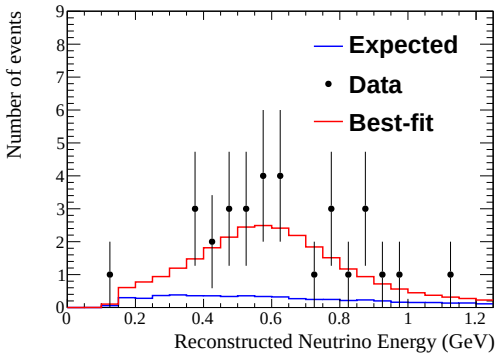
Table 24: The measured oscillation parameter best-fit and minimum χ^2 values as well as the 1σ CL intervals are shown, for both normal and inverted hierarchy. The $\pm1\sigma$ interval is obtained by using the *constant* $\Delta\chi^2$ method. All the parameters are measured using the reactor constraint $\sin^2 2\theta_{13} = 0.085 \pm 0.005$ [66], except for δ_{CP} and $\sin^2\theta_{13}$. As explained in section 4.5, the gaussian approximation does not work well for discrete parameters like the mass hierarchy however, since there is not a clear way to compute 1σ ranges for both mass hierarchies without performing CPU intensive toy MC studies, the constant $\Delta\chi^2$ method is used.

fixed to their best-fit values shown in table 24. Indeed these oscillation parameters have the strongest effect on the measurement, since $\sin^2\theta_{13}$ is constrained with the reactor measurement.

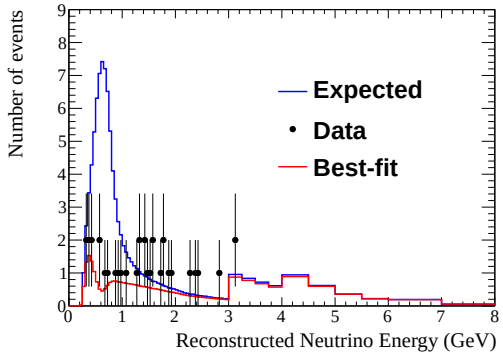
A goodness of test fit is performed by comparing data with the best fit spectra shown in fig. 27. In order to determine how compatible the data is with our model the log likelihood is calculated for each sample as in eq. (3) and the $\chi^2/ndof$ is used to estimate the fit probability using the Poisson statistic, given the number of degrees of freedom equal to $ndof = N_{bins} - N_{free}(8 - 5 = 3)$, where N_{bins} is the number of template bins and N_{free} is the number of free parameters in the fit. The free parameters are $\sin^2\theta_{23}$, δ_{CP} and the



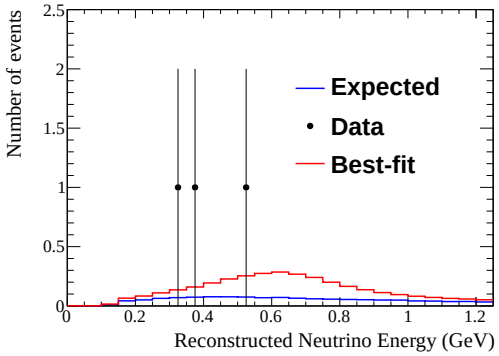
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode

Figure 27: The best-fit spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the run 1-6 data set. They correspond to the fit of $\sin^2\theta_{23}$, δ_{CP} and mass hierarchy with reactor constraint where all the nuisance parameters are marginalised. The method described in sec. 11 is used. The best-fit distributions are compared with the predicted unoscillated spectrum (all the oscillation parameters are set to 0), generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

mass hierarchy in this case. In this analysis, a coarse binning scheme is used for the goodness of fit test as in previous analyses. This is used so that there are enough events in each bin to calculate p-values in the gaussian approximation. 5 bins (0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV) are used for the neutrino mode muon-like spectrum, 1 bin is used for each of the muon-like antineutrino and electron-like spectra in order to enter the gaussian regime, where the χ^2 distribution is well defined. The χ^2 and resulting p-value for each spectrum is shown in Tab. 25. We see for all the spectra, the data and best fit spectra are very compatible.

	χ^2 / ndf	p-value
MH fixed	2.08917 / 6	0.84
MH fitted	2.08917 / 5	0.91

Table 25: Goodness of fit for the 2-dimensional fit of $\sin^2\theta_{23}$, δ_{CP} and mass hierarchy (both fitted and fixed) where all the other parameters are marginalised.

6.2. 2-dimensional confidence intervals

The same 2-dimensional confidence intervals as in sec. 5.1 are produced. In Fig. 28 the corresponding 2-dimensional contours produced with the Run 1-6 data set are shown drawing the normal and inverted hierarchy contours independently, i.e. each contour is produced with respect to the global minimum under that hierarchy hypothesis. In Fig. 29 both normal and inverted hierarchy contours are produced with respect to the same global minimum, i.e. the normal hierarchy hypothesis.

As already explained in section 4.5, "global" and "independent" contours correspond respectively to the least and most conservative approach.

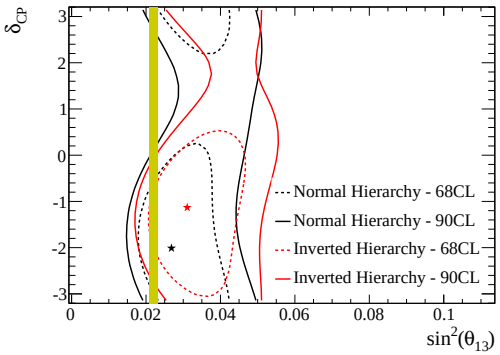
As in 1-dimensional analysis results, looking at contours of $\sin^2\theta_{13}$ Vs δ_{CP} with T2K only data, in Fig 28a, it is clear that the reactor measurements are in agreement with the T2K data at 1σ level if $\delta_{CP} \sim -\frac{\pi}{2}$. The tension between the $\sin^2\theta_{13}$ reactor constraint and the T2K data constraint is clear at $\sim \frac{+\pi}{2}$ where the T2K data prefer a lager mixing angle with both hierarchy hypotheses. This is the cause of the closed 90% contour in Fig 28b which is also in good agreement with the 1D δ_{CP} fit in Fig.23a. This effect is much greater for the inverted hierarchy fits and thus leads to a much tighter constraint on δ_{CP}. As a result, we see that all δ_{CP} constraints in Fig. 28b, 28c, 28d are tighter for the inverted hierarchy hypothesis than for the normal. Fig 28c also shows a slight anti-correlation between δ_{CP} and Δm^2_{32} in the normal hierarchy fit and a correlation in the inverted most likely coming from the fact that Δm^2_{32} and $cos(\delta_{CP})$ have similar effects on the shape of the appearance spectra. In Fig 28d we see correlation where the constraint on δ_{CP} is narrower for smaller values of $\sin^2\theta_{23}$ as they both appear in the appearance term. In Fig 28e we see that $\sin^2\theta_{23}$ and Δm^2_{32} are not very correlated however the inverted hierarchy contour is shifted down in Δm^2_{32} with respect to the normal hierarchy contour. This effect is due to the correlation between Δm^2_{32} and the mass hierarchy shown in Fig. 30. Indeed the inverted hierarchy hypothesis produces an event migration toward higher energy which is balanced in the fit by a lower fit value of Δm^2_{13} with respect to the normal hierarchy case.

The *constant* $\Delta\chi^2$ contours with respect to the global minimum in Fig. 29 show similar trends. As the normal hierarchy was preferred in all fits, the normal hierarchy contours are identical to those in Fig. 28. The inverted hierarchy contours are noticeably tighter in all cases. In Fig 29a we see that there is a closed $\Delta\chi^2 = 4.61$ contour, and in all others the $\Delta\chi^2 = 4.61$ contour is of similar size to the $\Delta\chi^2 = 2.3$ contour for the normal hierarchy.

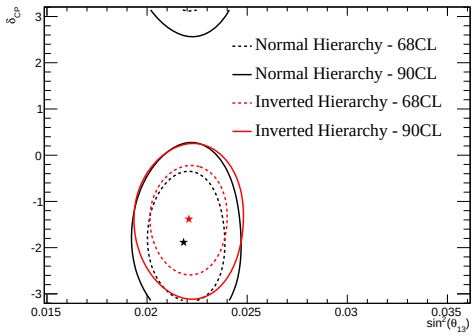
6.3. Discussion of the results

The value of δ_{CP} favored by the joint fit analysis when the reactor constraint is applied is -1.854 and corresponds to normal hierarchy. $\delta_{CP} = 0$ is excluded at 90% CL and a large fraction of the range between $\delta_{CP} = 0$ and $+\pi$ is excluded with a significance of 2σ (with the constant $\Delta\chi^2$ method). The inverted hierarchy is disfavored at more than $\sim 1\sigma$ level and it is partially excluded at 3σ for values of δ_{CP} around $+\pi/2$. It has been shown this exclusion, stronger than what is expected from the fit of the Asimov data sets for $\delta_{CP} = -1.601$, is due to the small excess of events in the neutrino mode e-like sample. Indeed the impact of antineutrino mode samples on the result is quite small. If the reactor constraint is not applied, the best-fit value of δ_{CP} is still quite close to $-\pi/2$. However the exclusion is much weaker than when the reactor constraint is applied. The contour of $\sin^2\theta_{13}$ Vs δ_{CP} shows an agreement at the level of 1σ with the measurement of the reactors if $\delta_{CP} \sim -\pi/2$.

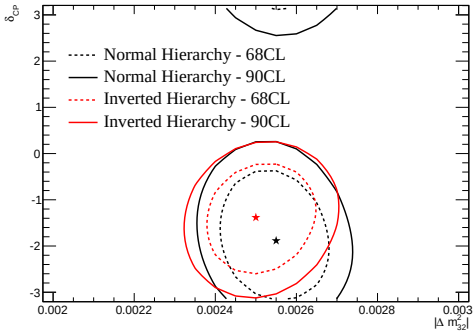
The result shows a stronger exclusion of δ_{CP} compared to the 2014 joint fit analysis [8]. This is mainly due to the new reactor constraint which has lower $\sin^2\theta_{13}$ (0.0217 instead of 0.0250) and a smaller uncertainty: 0.085 ± 0.005 instead of 0.095 ± 0.01.



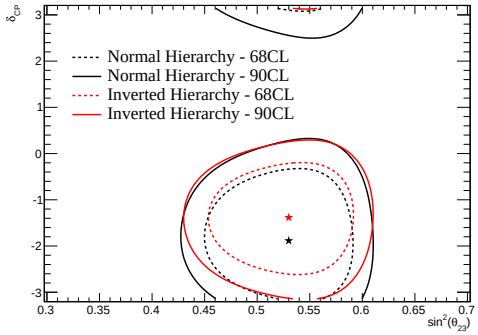
(a) $\sin^2\theta_{13}$ Vs δ_{CP} with T2K data only



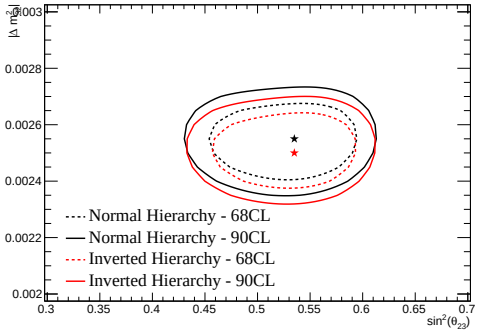
(b) $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint



(c) Δm^2_{32} (Δm^2_{13}) Vs δ_{CP} with reactor constraint

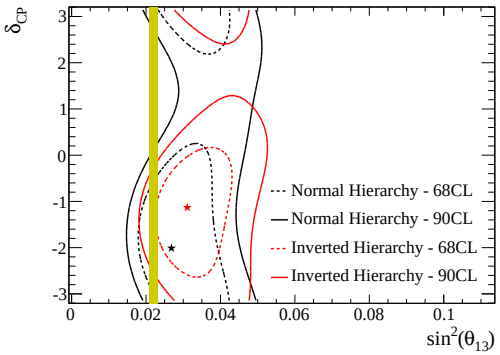


(d) $\sin^2\theta_{23}$ Vs δ_{CP} with reactor constraint

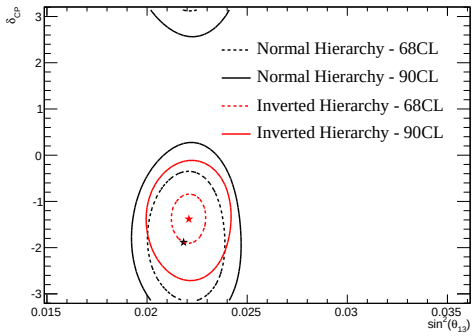


(e) $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with reactor constraint

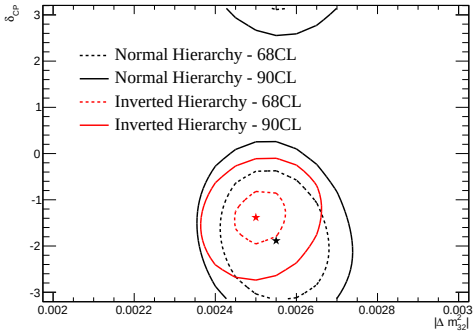
Figure 28: Measured confidence intervals with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed independently with respect to the global minimum of each hierarchy (star). All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured contour of $\sin^2\theta_{13}$ Vs δ_{CP} without reactor constraint, in Fig 28a is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.



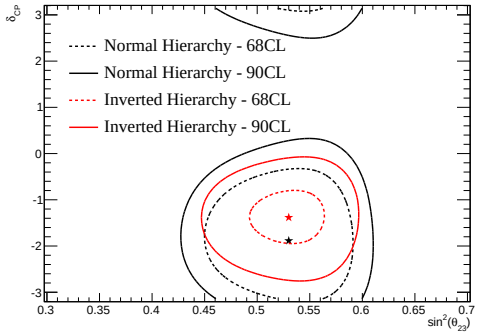
(a) $\sin^2\theta_{13}$ Vs δ_{CP} with T2K data only



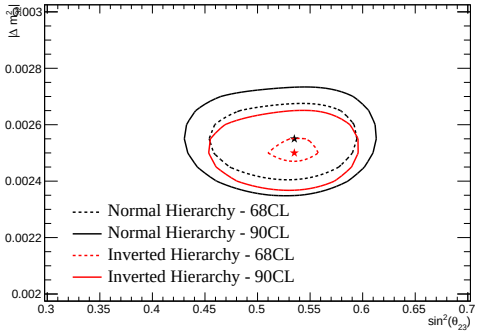
(b) $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint



(c) Δm^2_{32} (Δm^2_{13}) Vs δ_{CP} with reactor constraint

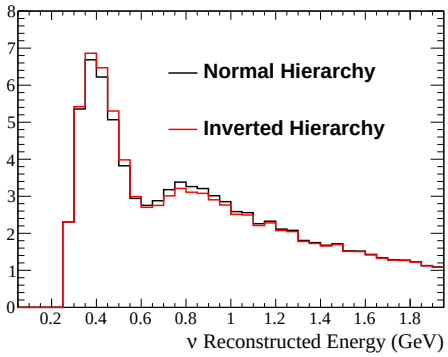


(d) $\sin^2\theta_{23}$ Vs δ_{CP} with reactor constraint

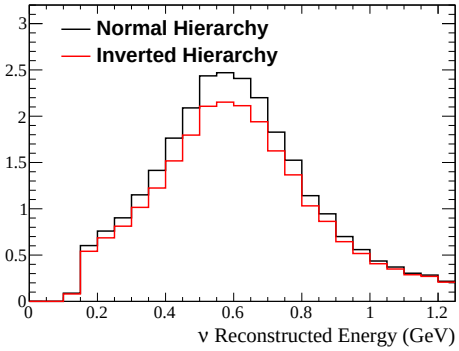


(e) $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with reactor constraint

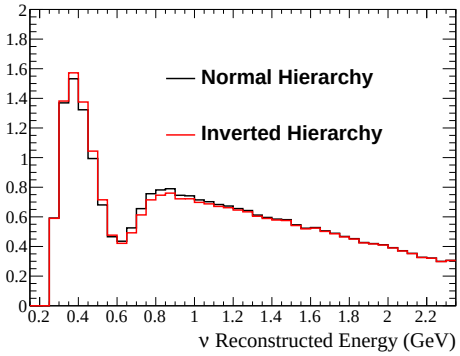
Figure 29: Measured confidence intervals with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. Contours for normal (black) and inverted (red) hierarchy are computed with respect to the minimum of the best-fit hierarchy (star), which corresponds to normal hierarchy. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured contour of $\sin^2\theta_{13}$ Vs δ_{CP} without reactor constraint is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.



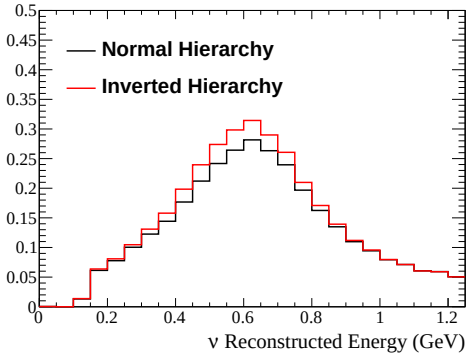
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode

Figure 30: The expected spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the run 1-6 data set for both normal (black) and inverted (red) hierarchies. They correspond to the Asimov data set A in table 12 and are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

7. Summary

In this technical note, we have presented the results of the 3-flavour $\bar{\nu}/\nu$ joint analysis performed by the *VALOR* group, on the combined Run 1-6 data set, for an exposure of 6.914×10^{20} POT in neutrino mode and 4.011×10^{20} POT in antineutrino mode. The goal of the analysis is to search for CP violation in the leptonic and put limits on δ_{CP} as well as measure all the other oscillation parameters to which T2K is sensitive: $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) and partially mass hierarchy, to which T2K has little sensitivity.

This analysis predicts 125.914 ± 6.459 (syst) for neutrino μ-like, 34.284 ± 1.755 (syst) for antineutrino μ-like, 26.594 ± 1.808 (syst) for neutrino e-like, and 3.218 ± 0.238 (syst) for antineutrino e-like single ring events in Super K and observes respectively 124 for neutrino μ-like, 34 for antineutrino μ-like, 31 for neutrino e-like, 3 for antineutrino e-like.

The $\bar{\nu}/\nu$ analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events and reconstructed energy versus lepton angle of single e-like ring events was used to compute the confidence intervals. All 101 systematic parameters as well as the oscillation parameters which were not parameters of interest considered in this analysis were marginalised.

From the fit of the Run 1-6 data we can see that the favored value of δ_{CP} is -1.854, close to $-\pi/2$. When the constant $\Delta\chi^2$ method is used $\delta_{CP} = 0$ is excluded at 90% CL and a large portion between 0 and $+\pi$ is excluded at 2σ if normal hierarchy is considered. In case of inverted hierarchy a not negligible range of δ_{CP} around $+\pi/2$ is excluded with a significance of 3σ.

From the contour of $\sin^2\theta_{13}$ Vs δ_{CP} without the reactor constraint it is clear that the T2K data are in agreement with the reactor measurements within 1σ if $\delta_{CP} \sim -\pi/2$.

It has also been shown that the result is dominated by the excess of event in the neutrino mode e-like sample, though the antineutrino mode e-like also prefers values of δ_{CP} close to $-\pi/2$, though the significance is not high.

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Appendices

A. Predicted number of events

In Tab. 26, 27 and 28, 29 the total number of events expected for the Asimov data set B (see table 12) is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	69.747	0.035	4.335	0.002	0.151	0.001	74.270
CC1pi	17.206	0.017	1.572	0.001	0.049	0.000	18.845
CCcoherent	0.330	0.000	0.080	0.000	0.001	0.000	0.412
MEC	19.725	0.010	0.852	0.000	0.036	0.000	20.623
CCother	7.742	0.008	0.494	0.001	0.002	0.000	8.247
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	122.243	0.297	7.659	0.025	0.238	0.002	130.464

Table 26: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	7.088	0.004	13.306	0.005	0.004	0.016	20.422
CC1pi	2.718	0.003	3.066	0.002	0.002	0.003	5.794
CCcoherent	0.054	0.000	0.173	0.000	0.000	0.000	0.228
MEC	2.525	0.001	2.118	0.001	0.001	0.002	4.649
CCother	1.282	0.001	0.886	0.000	0.000	0.000	2.170
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.637	0.051	20.282	0.032	0.008	0.022	35.032

Table 27: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.062	2.129	0.001	0.088	10.811	0.121	13.211
CC1pi	0.017	0.348	0.000	0.027	1.236	0.020	1.648
CCcoherent	0.000	0.002	0.000	0.002	0.011	0.002	0.017
MEC	0.009	0.710	0.000	0.022	2.750	0.022	3.513
CCother	0.001	0.045	0.000	0.003	0.025	0.001	0.076
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.476	3.264	0.069	0.145	14.833	0.167	19.954

Table 28: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.269	1.376	2.178
CC1pi	0.002	0.040	0.001	0.060	0.045	0.163	0.312
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.022	0.029
MEC	0.001	0.069	0.000	0.063	0.087	0.219	0.439
CCother	0.001	0.008	0.000	0.004	0.003	0.003	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.142	0.313	0.235	0.478	0.404	1.783	3.355

Table 29: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	338.683	0.037	8.672	0.002	0.000	0.000	347.394
CC1pi	39.668	0.018	2.276	0.001	0.000	0.000	41.963
CCcoherent	0.678	0.000	0.136	0.000	0.000	0.000	0.815
MEC	74.638	0.011	1.548	0.000	0.000	0.000	76.198
CCother	8.732	0.009	0.560	0.001	0.000	0.000	9.301
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	469.893	0.301	13.518	0.025	0.000	0.000	483.737

Table 30: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	15.420	0.004	55.525	0.005	0.000	0.000	70.954
CC1pi	4.071	0.003	6.536	0.002	0.000	0.000	10.611
CCcoherent	0.074	0.000	0.612	0.000	0.000	0.000	0.687
MEC	4.721	0.001	7.811	0.001	0.000	0.000	12.534
CCother	1.457	0.001	1.008	0.000	0.000	0.000	2.467
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	26.713	0.052	72.225	0.032	0.000	0.000	99.021

Table 31: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

991 In Tab. 30, 31 and 32, 33 the total number of unoscillated events is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.413	2.294	0.002	0.094	0.000	0.000	2.803
CC1pi	0.069	0.371	0.001	0.028	0.000	0.000	0.469
CCcoherent	0.002	0.003	0.000	0.002	0.000	0.000	0.006
MEC	0.095	0.759	0.000	0.023	0.000	0.000	0.878
CCother	0.004	0.046	0.000	0.003	0.000	0.000	0.054
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.970	3.504	0.071	0.154	0.000	0.000	5.698

Table 32: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.012	0.203	0.018	0.361	0.000	0.000	0.593
CC1pi	0.004	0.043	0.003	0.064	0.000	0.000	0.114
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.000	0.007
MEC	0.003	0.074	0.003	0.067	0.000	0.000	0.147
CCother	0.001	0.009	0.000	0.004	0.000	0.000	0.015
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.156	0.334	0.255	0.507	0.000	0.000	1.253

Table 33: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

B. Predicted *e*-like spectra

In Fig. 31 each flavor component of the neutrino and antineutrino mode e-like samples is shown separately in order to compare the different shapes as a function of E_{reco}. The NC spectrum distribution, which is the largest component of the background, is quite flat over all the energy range and is quite difficult to separate. The $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events populate the same energy region, around 0.6 GeV, with a very similar shape.

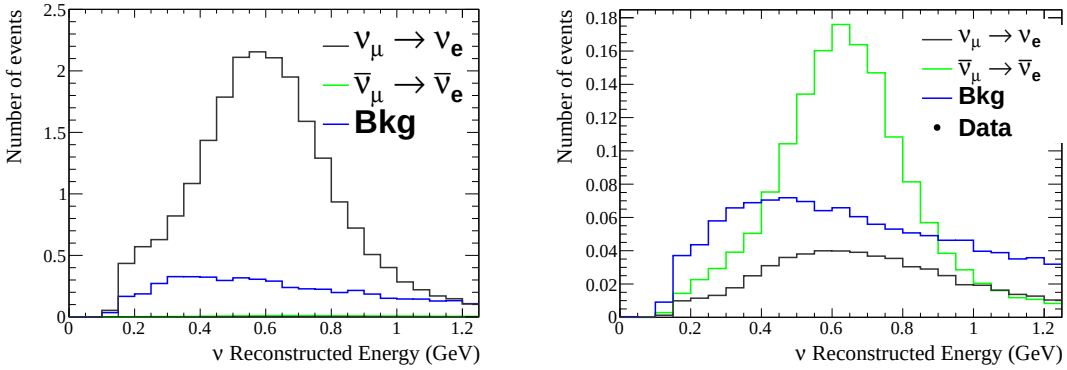


Figure 31: Predicted spectra of reconstructed neutrino energy for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode and of 4.011×10^{20} POT in antineutrino mode. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), background, which includes NC, $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

Another kinematical variable is thus needed in order to better separate different flavour / reaction compoonents. In Fig. 35 and 36 the 2-dimensional E_{reco} Vs θ spectra are shown for each flavour component and NC for both neutrino and antineutrino mode e-like samples. In particular the shapes of $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ distributions are quite different, thus this templates have a better capability to discriminate signal from background. In general this template parametrization produces quite different shapes for all the flavour and NC components.

In Fig. 34 the projection of the 2-dimensional e-like templates onto the E_{reco} and θ axes are shown.

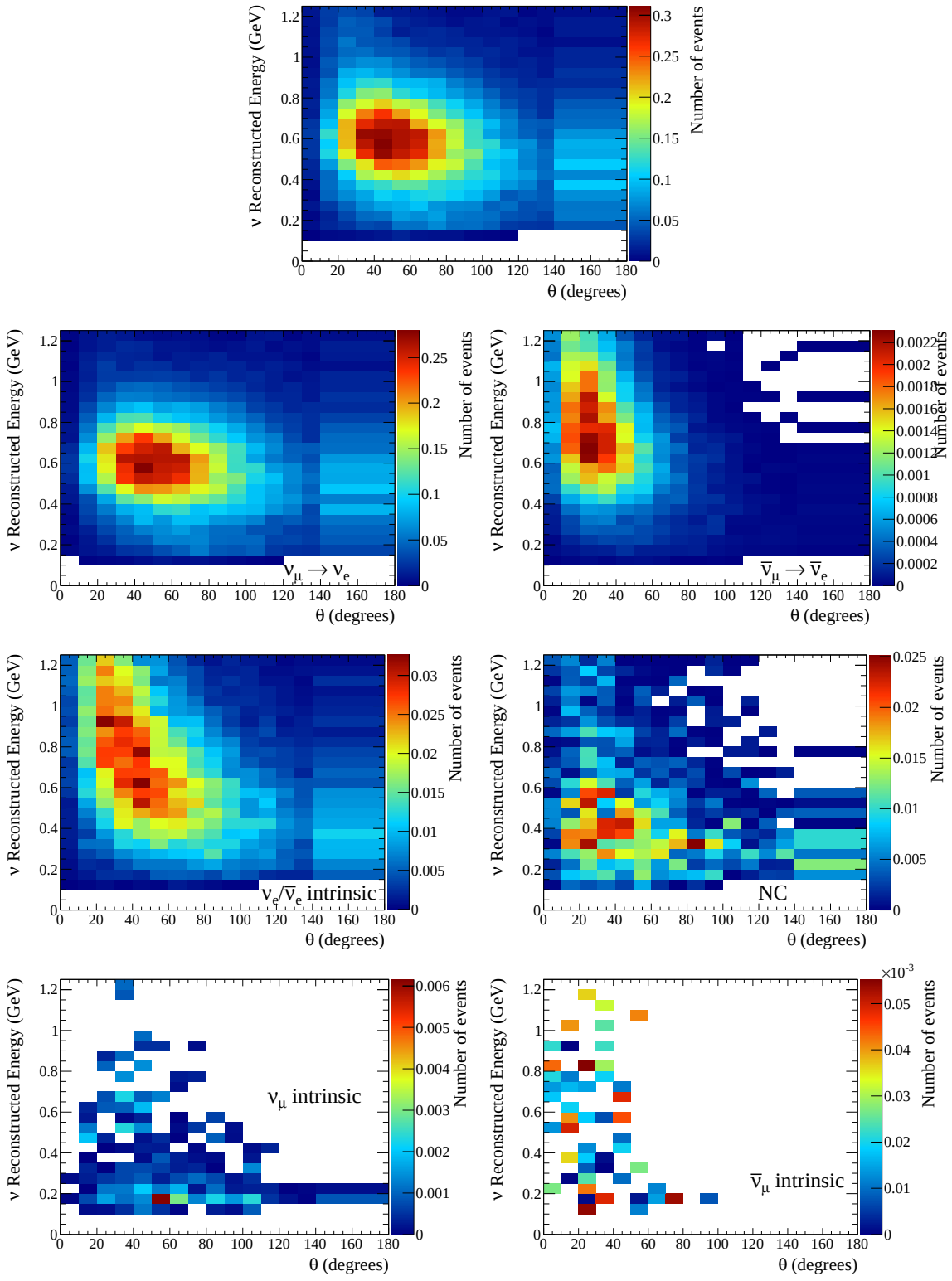


Figure 32: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

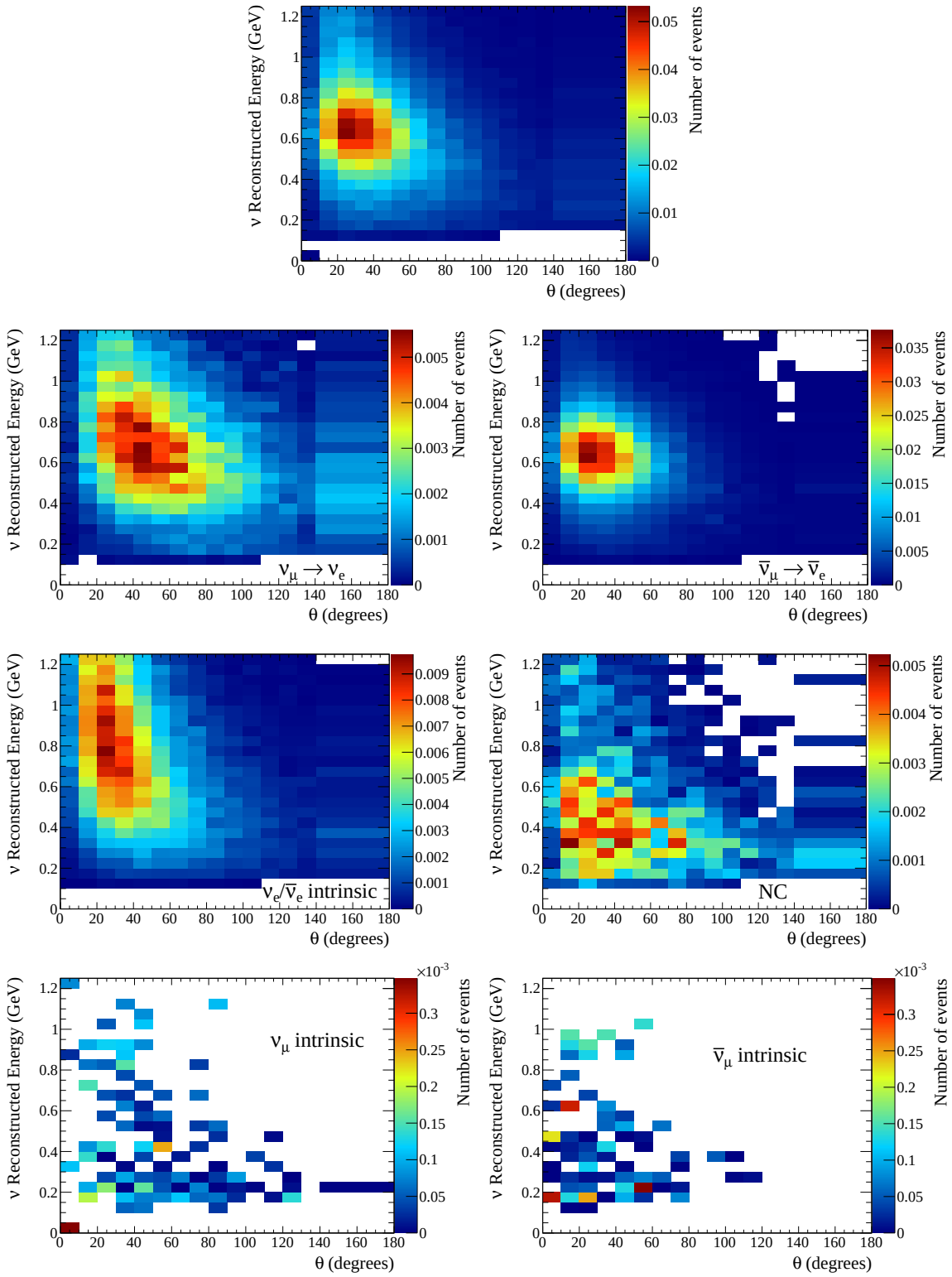


Figure 33: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of 4.011×10^{20} POT in antineutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

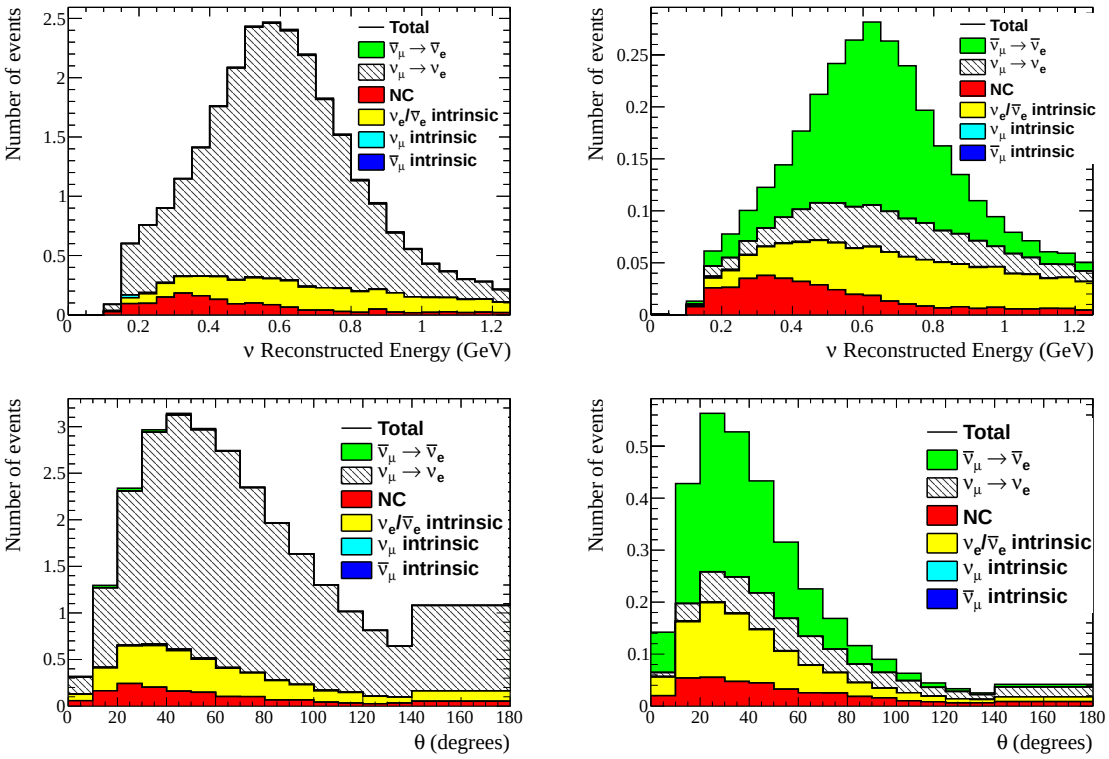


Figure 34: Predicted spectra of the reconstructed neutrino energy (top) and the angle between the neutrino and the outgoing lepton directions (bottom) for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode and of 4.011×10^{20} POT in antineutrino mode. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

C. Observed *e*-like spectra

In Fig. 35 and 36 the predicted e-like 2-dimensional spectra are compared to data for both neutrino and antineutrino beam mode.

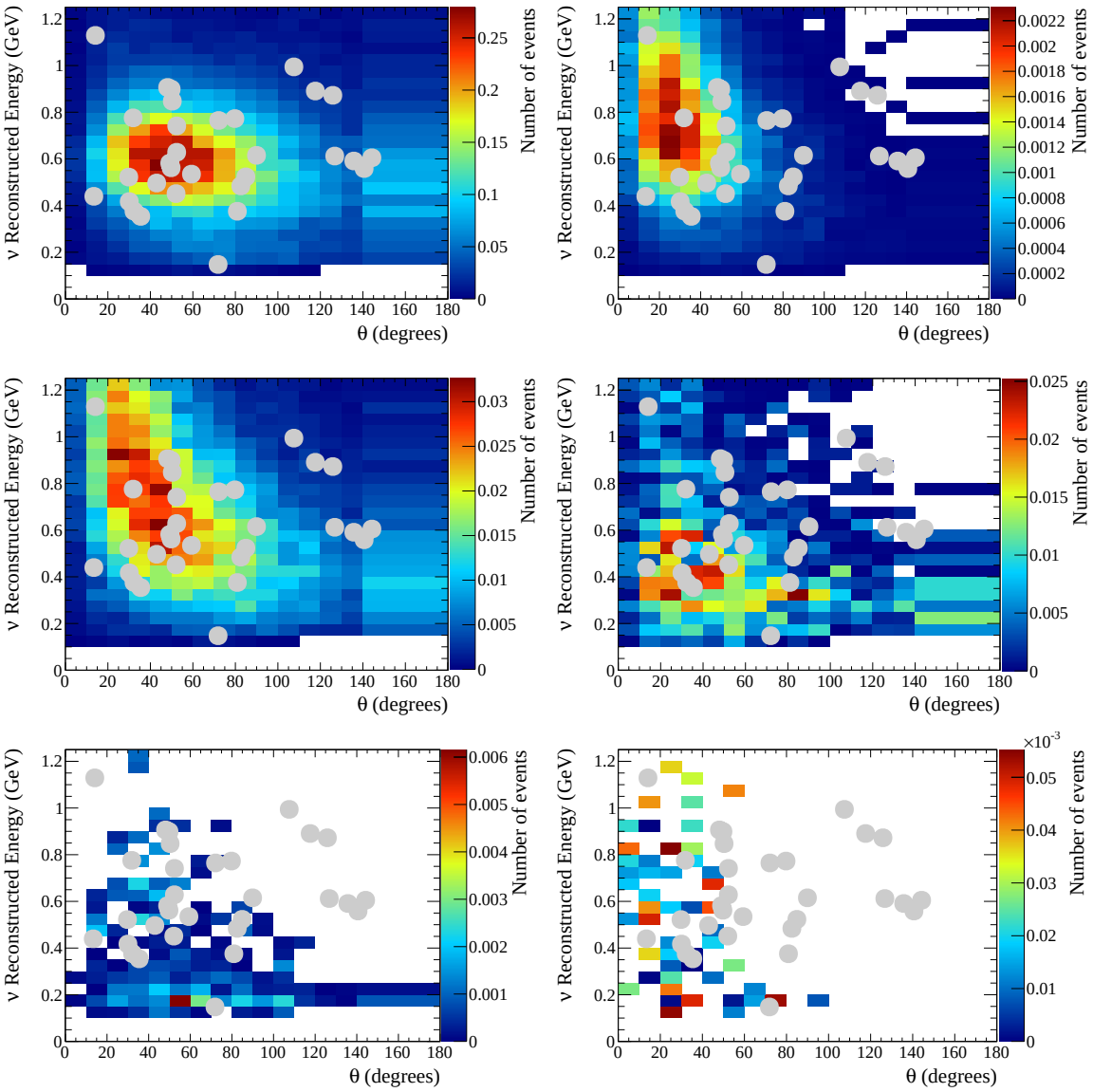


Figure 35: Predicted and observed (grey dots) spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

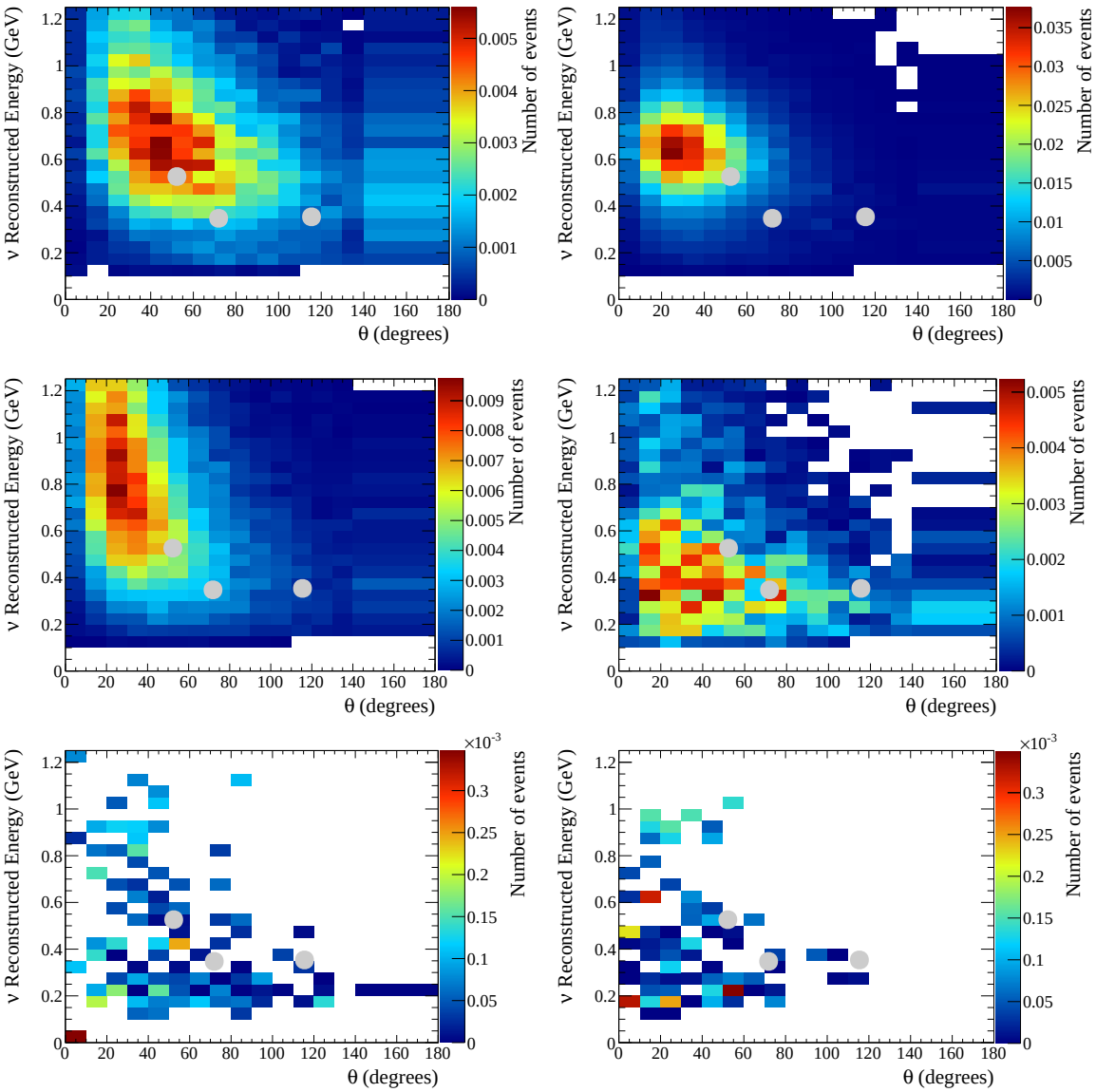


Figure 36: Predicted and observed (grey dots) spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of 4.011×10^{20} POT in antineutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

D. Effect of systematic uncertainties

In table 34, table 35, table 36 and table 35 it is shown which systematic parameter is applied to which reaction and flavour modes for all the Super-K samples.

	$\sin^2(\theta_{12})$	$\sin^2(\theta_{13})$	Δm^2_{21}	δ_{CP}	$f^{SK}_{E;r}$	$f^{SK+FSI}_{0;t,r}$	$f^{SK+FSI}_{1;t,r}$	$f^{SK+FSI}_{2;t,r}$	$f^{SK+FSI}_{3;t,r}$	$f^{SK+FSI}_{4;t,r}$	$f^{SK+FSI}_{5;t,r}$	$f^{banff}_{0;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{7;t,r}$	$f^{banff}_{8;t,r}$	$f^{banff}_{9;t,r}$	$f^{banff}_{10;t,r}$	$f^{banff}_{11;t,r}$	$f^{banff}_{12;t,r}$	$f^{banff}_{13;t,r}$
ν_μ CCQE	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC 1π	✓	✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC coherent	✓	✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC MEC	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC other	✓	✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC $1\pi^{+/-}$					✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC $1\pi^0$					✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC coherent					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC other					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_e NC 1γ					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
$\bar{\nu}_\mu$ CCQE					✓	✓			✓	✓													✓	✓	✓
$\bar{\nu}_\mu$ CC 1π					✓				✓														✓	✓	✓
$\bar{\nu}_\mu$ CC coherent					✓				✓														✓	✓	✓
$\bar{\nu}_\mu$ CC MEC					✓	✓	✓	✓															✓	✓	✓
$\bar{\nu}_\mu$ CC other					✓				✓														✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$					✓						✓												✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$					✓						✓												✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent					✓						✓												✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other					✓						✓												✓	✓	✓
ν_μ NC 1γ					✓						✓												✓	✓	✓
ν_e CCQE	✓	✓	✓	✓	✓						✓														
ν_e CC 1π	✓	✓	✓	✓	✓						✓														
ν_e CC coherent	✓	✓	✓	✓	✓						✓														
ν_e CC MEC	✓	✓	✓	✓	✓						✓														
ν_e CC other	✓	✓	✓	✓	✓						✓														
ν_e NC $1\pi^{+/-}$					✓						✓														
ν_e NC $1\pi^0$					✓						✓														
ν_e NC coherent					✓						✓														
ν_e NC other					✓						✓														
$\bar{\nu}_\mu$ NC 1γ					✓						✓														
Osc. ν_e CCQE	✓	✓	✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC 1π	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC coherent	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC MEC	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC other	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓						✓														
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓						✓														
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓						✓														
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓						✓														
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓						✓														
$\bar{\nu}_e$ NC other											✓														
$\bar{\nu}_e$ NC $1\pi^0$											✓														
$\bar{\nu}_e$ NC $1\pi^{+/-}$											✓														
$\bar{\nu}_e$ CC other										✓															
$\bar{\nu}_e$ CC MEC										✓															
$\bar{\nu}_e$ CC coherent										✓															
$\bar{\nu}_e$ CC 1π										✓															
$\bar{\nu}_e$ CCQE										✓															
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓					✓												✓	✓	✓	✓
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓					✓												✓	✓	✓	✓
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓					✓												✓	✓	✓	✓
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓					✓												✓	✓	✓	✓
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓					✓												✓	✓	✓	✓

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$f^{banff}_{7;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{0;t,r}$	$f^{SK+FSI}_{17;t,r}$	$f^{SK+FSI}_{16;t,r}$	$f^{SK+FSI}_{15;t,r}$	$f^{SK+FSI}_{14;t,r}$	$f^{SK+FSI}_{13;t,r}$	$f^{SK+FSI}_{12;t,r}$	$f^{SK+FSI}_{11;t,r}$	$f^{SK+FSI}_{10;t,r}$	$f^{SK+FSI}_{9;t,r}$	$f^{SK+FSI}_{8;t,r}$	$f^{SK+FSI}_{7;t,r}$	$f^{SK+FSI}_{6;t,r}$	$f^{SK}_{E;r}$	δ_{CP}	Δm^2_{21}	$\sin^2(\theta_{13})$	$\sin^2(\theta_{12})$	
✓	✓	✓	✓	✓	✓	✓	✓							✓	✓	✓				✓	✓	✓	✓	✓	ν_μ CCQE
✓	✓	✓	✓	✓	✓	✓	✓							✓	✓	✓				✓	✓	✓	✓	✓	ν_μ CC 1π
✓	✓	✓	✓	✓	✓	✓	✓							✓	✓	✓				✓	✓	✓	✓	✓	ν_μ CC coherent
✓	✓	✓	✓	✓	✓	✓	✓							✓	✓	✓				✓	✓	✓	✓	✓	ν_μ CC MEC
✓	✓	✓	✓	✓	✓	✓	✓							✓	✓	✓				✓	✓	✓	✓	✓	ν_μ CC other
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓										✓					ν_μ/ν_τ NC $1\pi^{+/-}$
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓										✓					ν_μ/ν_τ NC $1\pi^0$
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓										✓					ν_μ/ν_τ NC coherent
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓										✓					ν_μ/ν_τ NC other
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓										✓					ν_e NC 1γ
														✓	✓	✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CCQE
														✓	✓	✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC 1π
														✓	✓	✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC coherent
														✓	✓	✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC MEC
														✓	✓	✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC other
								✓	✓	✓										✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$
								✓	✓	✓										✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$
								✓	✓	✓										✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent
								✓	✓	✓										✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other
								✓	✓	✓										✓					ν_μ NC 1γ
											✓	✓	✓							✓	✓	✓	✓	✓	ν_e CCQE
											✓	✓	✓							✓	✓	✓	✓	✓	ν_e CC 1π
											✓	✓	✓							✓	✓	✓	✓	✓	ν_e CC coherent
											✓	✓	✓							✓	✓	✓	✓	✓	ν_e CC MEC
											✓	✓	✓							✓	✓	✓	✓	✓	ν_e CC other
								✓	✓	✓										✓					ν_e NC $1\pi^{+/-}$
								✓	✓	✓										✓					ν_e NC $1\pi^0$
								✓	✓	✓										✓					ν_e NC coherent
								✓	✓	✓										✓					ν_e NC other
								✓	✓	✓										✓					$\bar{\nu}_\mu$ NC 1γ
✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓	Osc. ν_e CCQE
✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓	Osc. ν_e CC 1π
✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓	Osc. ν_e CC coherent
✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓	Osc. ν_e CC MEC
✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓	Osc. ν_e CC other
											✓	✓	✓							✓	✓	✓	✓	✓	$\bar{\nu}_e$ CCQE
											✓	✓	✓							✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC 1π
											✓	✓	✓							✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC coherent
											✓	✓	✓							✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC MEC
											✓	✓	✓							✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC other
								✓	✓	✓										✓					$\bar{\nu}_e$ NC $1\pi^{+/-}$
								✓	✓	✓										✓					$\bar{\nu}_e$ NC $1\pi^0$
								✓	✓	✓										✓					$\bar{\nu}_e$ NC coherent
								✓	✓	✓										✓					$\bar{\nu}_e$ NC other
								✓	✓	✓										✓					$\bar{\nu}_e$ NC 1γ
																	✓	✓	✓	✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CCQE
																	✓	✓	✓	✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC 1π
																	✓	✓	✓	✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC coherent
																	✓	✓	✓	✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC MEC
																	✓	✓	✓	✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC other

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$f^{banff\ RHC}_{13;t,r}$	$f^{banff\ RHC}_{12;t,r}$	$f^{banff\ RHC}_{11;t,r}$	$f^{banff\ RHC}_{10;t,r}$	$f^{banff\ RHC}_{9;t,r}$	$f^{banff\ RHC}_{8;t,r}$	$f^{banff\ RHC}_{7;t,r}$	$f^{banff\ RHC}_{6;t,r}$	$f^{banff\ RHC}_{5;t,r}$	$f^{banff\ RHC}_{4;t,r}$	$f^{banff\ RHC}_{3;t,r}$	$f^{banff\ RHC}_{2;t,r}$	$f^{banff\ RHC}_{1;t,r}$	$f^{banff\ RHC}_{0;t,r}$	$f^{SK+FSI\ RHC}_{5;t,r}$	$f^{SK+FSI\ RHC}_{4;t,r}$	$f^{SK+FSI\ RHC}_{3;t,r}$	$f^{SK+FSI\ RHC}_{2;t,r}$	$f^{SK+FSI\ RHC}_{1;t,r}$	$f^{SK+FSI\ RHC}_{0;t,r}$	$f^{SK}_{E;r}$	δ_{CP}	Δm^2_{21}	$\sin^2(\theta_{13})$	$\sin^2(\theta_{12})$	
			✓	✓	✓	✓	✓				✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	ν_μ CCQE
			✓	✓	✓	✓	✓			✓			✓			✓				✓	✓	✓	✓	✓	ν_μ CC 1π
			✓	✓	✓	✓	✓			✓			✓			✓				✓	✓	✓	✓	✓	ν_μ CC coherent
			✓	✓	✓	✓	✓				✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	ν_μ CC MEC
			✓	✓	✓	✓	✓			✓			✓			✓				✓	✓	✓	✓	✓	ν_μ CC other
			✓	✓	✓	✓	✓	✓					✓	✓						✓					ν_μ/ν_τ NC $1\pi^{+/-}$
			✓	✓	✓	✓	✓	✓					✓	✓						✓					ν_μ/ν_τ NC $1\pi^0$
			✓	✓	✓	✓	✓	✓					✓	✓						✓					ν_μ/ν_τ NC coherent
			✓	✓	✓	✓	✓	✓					✓	✓						✓					ν_μ/ν_τ NC other
			✓	✓	✓	✓	✓	✓					✓	✓						✓					ν_e NC 1γ
✓	✓	✓									✓						✓	✓	✓	✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CCQE
✓	✓	✓														✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC 1π
✓	✓	✓														✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC coherent
✓	✓	✓									✓						✓	✓	✓	✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC MEC
✓	✓	✓														✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC other
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other
✓	✓	✓												✓						✓					ν_μ NC 1γ
															✓					✓	✓	✓	✓	✓	ν_e CCQE
															✓					✓	✓	✓	✓	✓	ν_e CC 1π
															✓					✓	✓	✓	✓	✓	ν_e CC coherent
															✓					✓	✓	✓	✓	✓	ν_e CC MEC
															✓					✓	✓	✓	✓	✓	ν_e CC other
														✓						✓					ν_e NC $1\pi^{+/-}$
														✓						✓					ν_e NC $1\pi^0$
														✓						✓					ν_e NC coherent
														✓						✓					ν_e NC other
														✓						✓					$\bar{\nu}_\mu$ NC 1γ
			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓					✓	✓	✓	✓	✓	Osc. ν_e CCQE
			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓					✓	✓	✓	✓	✓	Osc. ν_e CC 1π
			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓					✓	✓	✓	✓	✓	Osc. ν_e CC coherent
			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓					✓	✓	✓	✓	✓	Osc. ν_e CC MEC
			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓					✓	✓	✓	✓	✓	Osc. ν_e CC other
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CCQE
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC 1π
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC coherent
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC MEC
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC other
														✓						✓					$\bar{\nu}_e$ NC $1\pi^{+/-}$
														✓						✓					$\bar{\nu}_e$ NC $1\pi^0$
														✓						✓					$\bar{\nu}_e$ NC coherent
														✓						✓					$\bar{\nu}_e$ NC other
														✓						✓					$\bar{\nu}_e$ NC 1γ
✓	✓	✓													✓					✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CCQE
✓	✓	✓													✓					✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC 1π
✓	✓	✓													✓					✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC coherent
✓	✓	✓													✓					✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC MEC
✓	✓	✓													✓					✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC other

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	$\sin^2(\theta_{12})$	$\sin^2(\theta_{13})$	Δm^2_{21}	δ_{CP}	$f^{SK}_{E;r}$	$f^{SK+FSI}_{6;t,r}\ RHC$	$f^{SK+FSI}_{7;t,r}\ RHC$	$f^{SK+FSI}_{8;t,r}\ RHC$	$f^{SK+FSI}_{9;t,r}\ RHC$	$f^{SK+FSI}_{10;t,r}\ RHC$	$f^{SK+FSI}_{11;t,r}\ RHC$	$f^{SK+FSI}_{12;t,r}\ RHC$	$f^{SK+FSI}_{13;t,r}\ RHC$	$f^{SK+FSI}_{14;t,r}\ RHC$	$f^{SK+FSI}_{15;t,r}\ RHC$	$f^{SK+FSI}_{16;t,r}\ RHC$	$f^{SK+FSI}_{17;t,r}\ RHC$	$f^{banff}_{0;t,r}\ RHC$	$f^{banff}_{1;t,r}\ RHC$	$f^{banff}_{2;t,r}\ RHC$	$f^{banff}_{3;t,r}\ RHC$	$f^{banff}_{4;t,r}\ RHC$	$f^{banff}_{5;t,r}\ RHC$	$f^{banff}_{6;t,r}\ RHC$	$f^{banff}_{7;t,r}\ RHC$
ν_μ CCQE	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC 1π	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC coherent	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC MEC	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC other	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC $1\pi^{+/-}$					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC $1\pi^0$					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC coherent					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC other					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_e NC 1γ					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CCQE					✓				✓	✓	✓													
$\bar{\nu}_\mu$ CC 1π					✓				✓	✓	✓													
$\bar{\nu}_\mu$ CC coherent					✓				✓	✓	✓													
$\bar{\nu}_\mu$ CC MEC					✓				✓	✓	✓													
$\bar{\nu}_\mu$ CC other					✓				✓	✓	✓													
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$					✓										✓	✓	✓							
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$					✓										✓	✓	✓							
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent					✓										✓	✓	✓							
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other					✓										✓	✓	✓							
ν_μ NC 1γ					✓										✓	✓	✓							
ν_e CCQE	✓	✓	✓	✓	✓							✓	✓	✓										
ν_e CC 1π	✓	✓	✓	✓	✓							✓	✓	✓										
ν_e CC coherent	✓	✓	✓	✓	✓							✓	✓	✓										
ν_e CC MEC	✓	✓	✓	✓	✓							✓	✓	✓										
ν_e CC other	✓	✓	✓	✓	✓							✓	✓	✓										
ν_e NC $1\pi^{+/-}$					✓										✓	✓	✓							
ν_e NC $1\pi^0$					✓										✓	✓	✓							
ν_e NC coherent					✓										✓	✓	✓							
ν_e NC other					✓										✓	✓	✓							
$\bar{\nu}_\mu$ NC 1γ					✓										✓	✓	✓							
Osc. ν_e CCQE	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC 1π	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC coherent	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC MEC	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC other	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓							✓	✓	✓										
$\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓							✓	✓	✓										
$\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓							✓	✓	✓										
$\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓							✓	✓	✓										
$\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓							✓	✓	✓										
$\bar{\nu}_e$ NC $1\pi^{+/-}$					✓										✓	✓	✓							
$\bar{\nu}_e$ NC $1\pi^0$					✓										✓	✓	✓							
$\bar{\nu}_e$ NC coherent					✓										✓	✓	✓							
$\bar{\nu}_e$ NC other					✓										✓	✓	✓							
$\bar{\nu}_e$ NC 1γ					✓										✓	✓	✓							
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓	✓	✓	✓																

Continued on next page

Table 37 – *Continued from previous page*

$f^{banff}_{8;t,r}$ RHC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓	✓													
$f^{banff}_{9;t,r}$ RHC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓	✓													
$f^{banff}_{10;t,r}$ RHC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓	✓													
$f^{banff}_{11;t,r}$ RHC											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f^{banff}_{12;t,r}$ RHC											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f^{banff}_{13;t,r}$ RHC											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f^{banff}_{14;t,r}$ RHC											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f^{banff}_{15;t,r}$ RHC											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f^{banff}_{16;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{17;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{18;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{19;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{20;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{21;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{22;t,r}$ RHC																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																
$f^{banff}_{23;t,r}$ RHC																															✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
$f^{banff}_{24;t,r}$ RHC																															✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
$f^{banff}_{M^{QE}_A}$ RHC	✓										✓										✓								✓										✓							
$f^{banff}_{M^{RES}_A}$ RHC		✓				✓	✓					✓				✓	✓					✓				✓	✓																		
$f^{banff}_{Norm_{MEC}}$ RHC				✓										✓									✓																						
$f^{banff}_{C^A_5}$ RHC		✓				✓	✓					✓				✓	✓					✓	✓							✓	✓														
$f^{banff}_{Bg_{RES}}$ RHC		✓				✓	✓					✓				✓	✓					✓	✓							✓	✓														
$f^{banff}_{p_f}$ RHC	✓										✓										✓								✓										✓							
$f^{banff}_{Shape_{CCoth}}$ RHC					✓										✓										✓								✓											✓
$f^{banff}_{E_B}$ RHC	✓										✓										✓								✓										✓							
$f^{banff}_{Norm_{CCcoh}}$ RHC			✓										✓									✓								✓					✓						✓				
$f^{banff}_{Norm_{NCcoh}}$ RHC							✓										✓							✓												✓								
$f^{banff}_{Norm_{NCother}}$ RHC								✓										✓							✓												✓								
$f^{banff}_{Norm_{\nu_e \to \nu_\mu}}$ RHC																			✓	✓	✓	✓	✓					✓	✓	✓	✓	✓													

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Table 37 – *Continued from previous page*

| | | |

In tab. 38, 39, 40, 41 and 42 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set B.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	4.02%
Flux	3.56%
XSec Corr Tot	4.03%
MEC (corr)	3.49%
MECbar (corr)	0.20%
NC other (uncorr)	0.77%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.10%
Flux+XSec	2.85%
Flux+XSec+SKDet+FSI+SI	4.96%
Oscillations	0.10%
All	4.96%

Table 38: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	3.67%
Flux	3.61%
XSec Corr Tot	4.16%
MEC (corr)	3.94%
MECbar (corr)	0.07%
NC other (uncorr)	0.21%
NC 1gamma (uncorr)	1.94%
XSec nue/numu	2.55%
XSec Tot	5.26%
Flux+XSec	4.29%
Flux+XSec+SKDet+FSI+SI	5.68%
Oscillations	4.18%
All	7.19%

Table 39: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

In fig. 37 and 38 respectively the absolute error envelope and the fractional error envelopes expressed as a ratio of event count are shown for the Asimov data set B (see table 12).

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B
SKDet+FSI+SI	3.83%
Flux	3.70%
XSec Corr Tot	4.10%
MEC (corr)	2.95%
MECbar (corr)	1.83%
NC other (uncorr)	0.73%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.17%
Flux+XSec	3.31%
Flux+XSec+SKDet+FSI+SI	5.04%
Oscillations	0.06%
All	5.11%

Table 40: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	3.88%
Flux	3.81%
XSec Corr Tot	4.52%
MEC (corr)	2.92%
MECbar (corr)	2.56%
NC other (uncorr)	0.32%
NC 1gamma (uncorr)	2.81%
XSec nue/numu	1.68%
XSec Tot	5.59%
Flux+XSec	4.62%
Flux+XSec+SKDet+FSI+SI	6.18%
Oscillations	3.59%
All	7.09%

Table 41: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	μ-like RMS/Mean	e-like RMS/Mean
SKDet	0.6%	1.2%
FSI+SI	2.5%	3.6%
SKDet+FSI+SI	2.6%	3.7%
Flux	1.6%	1.7%
XSec NC other (uncorr)	0.0%	0.1%
XSec NC 1γ (uncorr)	0.0%	0.9%
XSec ν_e / ν_μ (uncorr)	0.0%	3.3%
XSec Tot	1.7%	4.4%
Flux+XSec	1.8%	4.1%
Flux+XSec+SDet+FSI+SI	3.1%	5.6%
Oscillations	0.0%	0.6%
All	3.2%	5.6%

Table 42: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

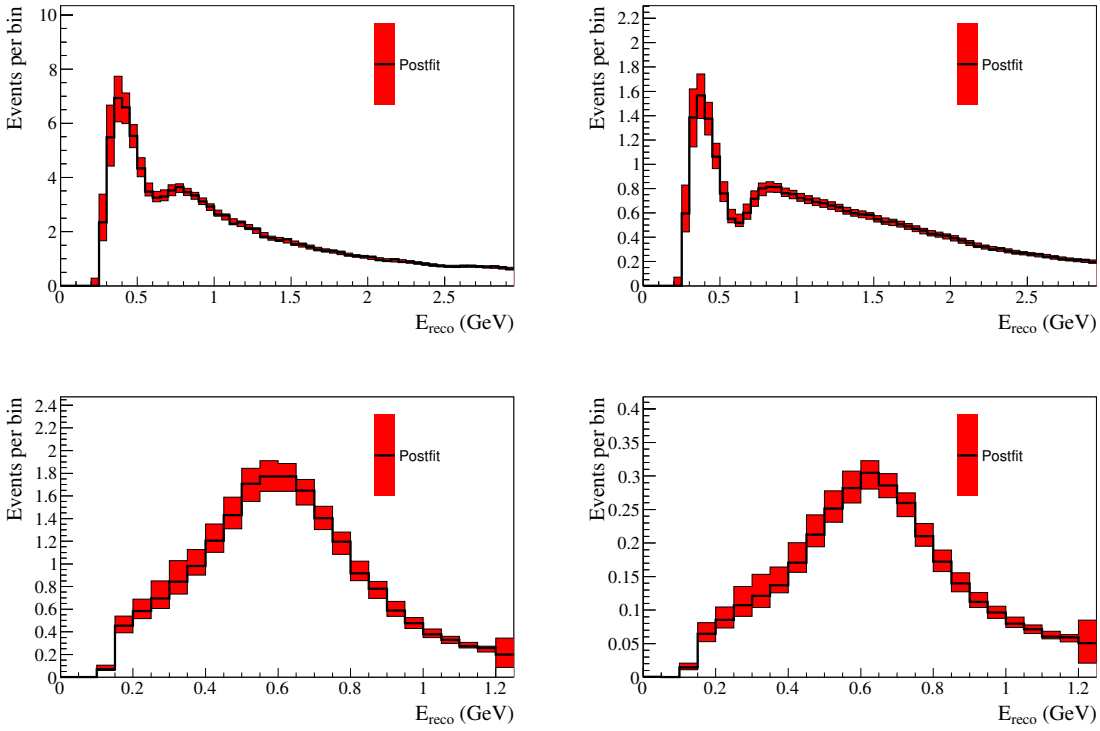


Figure 37: Error envelopes for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set B shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

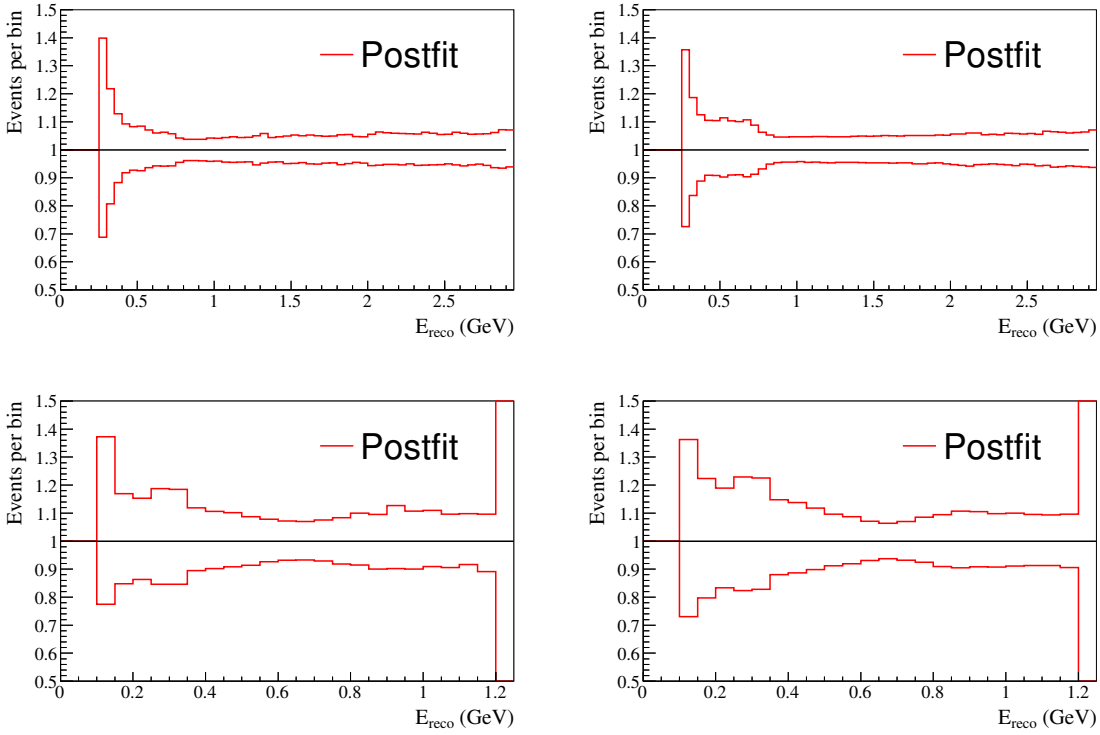


Figure 38: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set B shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

In tab. 43, 44, 45 and 46 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set A, but without taking into account the correction of the post-fit BANFF systematic parameters as well the additional covariance matrix that take into account the bias in the flux parameters of the BANFF fit. The impact of the uncertainty due to the bias of the BANFF fit is small and has a small impact compared to the other systematic uncertainties.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	4.11%
Flux	3.26%
XSec Corr Tot	3.96%
MEC (corr)	3.53%
MECbar (corr)	0.20%
NC other (uncorr)	0.79%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.04%
Flux+XSec	2.75%
Flux+XSec+SKDet+FSI+SI	5.03%
Oscillations	0.03%
All	5.00%

Table 43: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.47%
Flux	3.40%
XSec Corr Tot	4.17%
MEC (corr)	3.95%
MECbar (corr)	0.05%
NC other (uncorr)	0.16%
NC 1gamma (uncorr)	1.49%
XSec nue/numu	2.63%
XSec Tot	5.15%
Flux+XSec	4.08%
Flux+XSec+SKDet+FSI+SI	5.51%
Oscillations	4.19%
All	6.93%

Table 44: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.40%
XSec Corr Tot	4.20%
MEC (corr)	3.00%
MECbar (corr)	1.76%
NC other (uncorr)	0.76%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.27%
Flux+XSec	2.90%
Flux+XSec+SKDet+FSI+SI	4.90%
Oscillations	0.03%
All	4.91%

Table 45: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.53%
XSec Corr Tot	4.44%
MEC (corr)	3.11%
MECbar (corr)	2.24%
NC other (uncorr)	0.33%
NC 1gamma (uncorr)	2.98%
XSec nue/numu	1.52%
XSec Tot	5.57%
Flux+XSec	4.39%
Flux+XSec+SKDet+FSI+SI	6.11%
Oscillations	3.98%
All	7.24%

Table 46: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.